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CONVERGENCE OF THE LAX-FRIEDRICHS SCHEME FOR ISENTROPIC GAS DYNAMICS (II)*

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Abstract

A convergence theorem of the approximate solutions generated by the *Lax-Friedrichs* scheme for the isentropic equations of gas dynamics ($\gamma = \frac{3}{2}$) is established by using an analysis of the weak entropy and studying the regularity of the parametrized probability measures.

§ 1. Introduction

We shall deal with the convergence of the approximate solutions satisfying the compactness frame^[1], esp. generated by the *Lax-Friedrichs* scheme, of the *Cauchy* problem for the isentropic equations of gas dynamics

$$\left. \begin{aligned} \rho_t + (\rho u)_x &= 0, \\ (\rho u)_t + (\rho u^2 + p)_x &= 0, \quad p(\rho) = \text{const. } \rho^\gamma. \end{aligned} \right\} \quad (1.1)$$

$$(\rho, u)|_{t=0} = (\rho_0(x), u_0(x)), \quad (1.2)$$

In the context of hyperbolic system of conservation laws, the *Helly* principle is used to guarantee the existence of a subsequence converging pointwise a.e.. This principle asks a sequence to satisfy the following frame: the spatial L^∞ and total variation norms have a uniform control. *Oleinik*^[2], *Conway* and *Smoller*^[3] proved that the *Lax-Friedrichs* difference approximations for scalar conservation law satisfy the *Helly* frame.

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For general hyperbolic system, it is very difficult to prove that a sequence of approximate solutions, particularly the *Lax-Friedrichs* difference approximations, satisfies the frame. This motivates us to find a new compactness frame which is satisfied by the *Lax-Friedrichs* difference approximations and still guarantees the existence of a subsequence converging pointwise a.e. In the preceding paper^[1] we have established a compactness frame satisfied by the *Lax-Friedrichs* difference approximations for the *Cauchy* problem (1.1)–(1.2). For $\gamma = 1 + \frac{2}{2m+1}$, $m \geq 2$ integers, the convergence of the *Lax-Friedrichs* difference approximations can be verified by using the result of the paper^[4]. In this paper, for $\gamma = \frac{3}{2}$, we prove that the frame can guarantee the existence of a subsequence converging pointwise a.e. by a detailed analysis of the weak entropy and studying the regularity of the parametrized probability measures. As a direct corollary, the convergence of the *Lax-Friedrichs* difference approximations for the *Cauchy* problem (1.1)–(1.2) ($\gamma = \frac{3}{2}$) is obtained. For the general case $1 < \gamma \leq \frac{5}{3}$, it will be presented in the forthcoming paper (III).

It is necessary to indicate that the convergence theorem established in this paper is true for any approximate solutions satisfying the compactness frame^[1] for the isentropic equations of gas dynamics.

In this paper we assume that the reader is familiar with the arguments of [4].

§ 2. Weak Entropy

The weak entropy of system (1.1) ($\gamma = \frac{3}{2}$) has the following general expression, i.e. *Darboux* formula^[4]:

$$\eta(w, z) = \int_s^w [(w-s)(s-z)]^{\frac{3}{2}} \varphi(s) ds. \quad (2.1)$$

First of all, we study some weak entropies and limit functions of the weak entropy to meet the needs of the following section.

A. From *Darboux* formula (2.1), we may form a sequence of the weak entropies

$$\eta_n(w, z) = \int_s^w [(w-s)(s-z)]^{\frac{3}{2}} \varphi_n(s) ds = \langle \varphi_n(s), K_1(s; w, z) \rangle,$$

where

$$K_1(s; w, z) = \begin{cases} [(w-s)(s-z)]^{\frac{3}{2}}, & z \leq s \leq w, \\ 0, & \text{otherwise.} \end{cases}$$

1. Set $\{\varphi_n\}$ to be a sequence of smooth functions which approach the δ -function concentrated at the origin and which have compact support, then

$$\begin{aligned} \eta_n(w, z) &= \langle \varphi_n(s), K_1(s; w, z) \rangle \rightarrow \langle \delta_0(s), K_1(s; w, z) \rangle \\ &= (-zw)^{\frac{3}{2}} X(w, z) \equiv \eta_0(w, z), \end{aligned}$$

where

$$X(w, z) = \begin{cases} 1, & w > 0, z < 0, \\ 0, & \text{otherwise.} \end{cases}$$

Remark that in general the flux q of a weak pair is computed by

$$q = \lambda_2 \eta + \tau, \quad \tau = -\frac{5}{8} \int_0^w \eta(x, z) dx,$$

or

$$q = \lambda_1 \eta + \bar{\tau}, \quad \bar{\tau} = \frac{5}{8} \int_0^w \eta(w, y) dy.$$

Then

$$q_0 = \lambda_2 \eta_0 + \tau_0, \quad \tau_0 = -\frac{1}{4} (-zw)^{\frac{3}{2}} w X(w, z).$$

2. Similarly, set $\{\varphi_n\}$ to be a sequence of smooth functions which approach the derivative of $\frac{2}{3} \delta$ -function concentrated at the origin and which have compact support, then

$$\begin{aligned} \eta_n(w, z) &= \langle \varphi_n(s), K_1(s; w, z) \rangle \rightarrow \left\langle \frac{2}{3} \delta_0^{(1)}(s), K_1(s; w, z) \right\rangle \\ &= (-zw)^{\frac{1}{2}} (w+z) X(w, z) \equiv \eta_1(w, z). \end{aligned}$$

It follows that

$$q_1 = \lambda_2 \eta_1 + \tau_1, \quad \tau_1 = -\frac{1}{4} (-zw)^{\frac{1}{2}} w \left(w + \frac{5}{3} z \right) X(w, z).$$

3. For (η_0, q_0) and (η_1, q_1) , we have

$$\eta_0 q_1 - \eta_1 q_0 = \eta_0 \tau_1 - \eta_1 \tau_0 = \frac{1}{6} (-zw)^3 X(w, z) \geq 0. \quad (2.2)$$

B. For any $\psi(t) \in C_0^\infty(-1, 1)$, we replace $\varphi(s)$ by

$$\frac{d^2}{ds^2} \int_{-1}^s (s-t)^{-\frac{1}{2}} \psi'(t) dt$$

in (2.1) and have

$$\eta(w, z) = \int_s^w [(w-s)(s-z)]^{\frac{3}{2}} \frac{d^2}{ds^2} \int_{-1}^s (s-t)^{-\frac{1}{2}} \psi'(t) dt ds$$

$$\begin{aligned}
&= \int_z^w \frac{d^2[(w-s)(s-z)]^{\frac{3}{2}}}{ds^2} \int_{-1}^s (s-t)^{-\frac{1}{2}} \psi'(t) dt ds \\
&= \int_z^w \frac{d^2[(w-s)(s-z)]^{\frac{3}{2}}}{ds^2} \\
&\quad \times \lim_{\varepsilon \rightarrow 0} \left(\int_{-1}^{z-\varepsilon} + \int_{z+\varepsilon}^s \right) (s-t)^{-\frac{1}{2}} S(s-t) \psi'(t) dt ds,
\end{aligned}$$

where $S(x) = \begin{cases} 1, & x \geq 0, \\ 0, & x < 0. \end{cases}$

Notice that the limit is uniformly convergent with respect to s , then

$$\begin{aligned}
\eta(w, z) &= \lim_{\varepsilon \rightarrow 0} \left(\int_z^w ds \int_{-1}^{z-\varepsilon} + \int_{z+\varepsilon}^w ds \int_{z+\varepsilon}^s \right) \frac{d^2[(w-s)(s-z)]^{\frac{3}{2}}}{ds^2} (s-t)^{-\frac{1}{2}} \psi'(t) dt \\
&= \lim_{\varepsilon \rightarrow 0} \left\{ \int_{-1}^{z-\varepsilon} \psi'(t) f(t) dt + \int_{z+\varepsilon}^w \psi'(t) g(t) dt \right\} \\
&= \lim_{\varepsilon \rightarrow 0} \left\{ \psi(z-\varepsilon) f(z-\varepsilon) - \psi(z+\varepsilon) g(z+\varepsilon) + \psi(w) g(w) \right. \\
&\quad \left. - \int_{-1}^{z-\varepsilon} \psi(t) f'(t) dt - \int_{z+\varepsilon}^w \psi(t) g'(t) dt \right\}, \tag{2.3}
\end{aligned}$$

where

$$f(t) = \int_z^w \frac{d^2[(w-s)(s-z)]^{\frac{3}{2}}}{ds^2} (s-t)^{-\frac{1}{2}} ds, \quad -1 < t \leq z - \varepsilon,$$

and

$$g(t) = \int_t^w \frac{d^2[(w-s)(s-z)]^{\frac{3}{2}}}{ds^2} (s-t)^{-\frac{1}{2}} ds, \quad z + \varepsilon \leq t \leq w.$$

It is easy to compute

$$\begin{aligned}
f(z-\varepsilon) &= \frac{3}{4} \left\{ f_0(z-\varepsilon) + (w-z)^{\frac{3}{2}} \ln \frac{2(w-z) + \varepsilon + 2\sqrt{(w-z)(w-z+\varepsilon)}}{\varepsilon} \right\}, \\
g(z+\varepsilon) &= \frac{3}{4} \left\{ g_0(z+\varepsilon) + (w-z)^{\frac{3}{2}} \ln \frac{2(w-z) - \varepsilon + 2\sqrt{(w-z)(w-z-\varepsilon)}}{\varepsilon} \right\},
\end{aligned}$$

where

$$\begin{aligned}
f_0(z-\varepsilon) &= \int_z^w (w-s)^{-\frac{1}{2}} (s-z)^{\frac{3}{2}} (s-z+\varepsilon)^{-\frac{1}{2}} ds \\
&\quad - 6 \int_z^w (w-s)^{\frac{1}{2}} (s-z)^{\frac{1}{2}} (s-z+\varepsilon)^{-\frac{1}{2}} ds \\
&\quad - \frac{3}{2} \int_z^w (s-z)^{\frac{1}{2}} (s-z+\varepsilon)^{-\frac{1}{2}} ds \int_0^1 (w-s+\theta(s-z))^{\frac{1}{2}} d\theta, \\
g_0(z+\varepsilon) &= \int_{z+\varepsilon}^w (w-s)^{-\frac{1}{2}} (s-z)^{\frac{3}{2}} (s-z-\varepsilon)^{-\frac{1}{2}} ds \\
&\quad - 6 \int_{z+\varepsilon}^w (w-s)^{\frac{1}{2}} (s-z)^{\frac{1}{2}} (s-z-\varepsilon)^{-\frac{1}{2}} ds \\
&\quad - \frac{3}{2} \int_{z+\varepsilon}^w (s-z)^{\frac{1}{2}} (s-z-\varepsilon)^{-\frac{1}{2}} ds \int_0^1 (w-s+\theta(s-z))^{\frac{1}{2}} d\theta.
\end{aligned}$$

Using the control convergent theorem, we immediately obtain

$$\lim_{\varepsilon \rightarrow 0} (f_0(z-\varepsilon) - g_0(z+\varepsilon)) = 0.$$

Hence

$$\begin{aligned} & \lim_{\varepsilon \rightarrow 0} [\psi(z-\varepsilon)f(z-\varepsilon) - \psi(z+\varepsilon)g(z+\varepsilon)] \\ &= -\frac{1}{4} \left\{ (w-z)^{\frac{3}{2}} \lim_{\varepsilon \rightarrow 0} \left[\psi(z-\varepsilon) \ln \frac{2(w-z)+\varepsilon+2\sqrt{(w-z)(w-z+\varepsilon)}}{2(w-z)-\varepsilon+2\sqrt{(w-z)(w-z-\varepsilon)}} \right] \right. \\ & \quad \left. + \frac{\psi(z-\varepsilon)-\psi(z+\varepsilon)}{\varepsilon} \cdot \varepsilon \ln \frac{2(w-z)-\varepsilon+2\sqrt{(w-z)(w-z-\varepsilon)}}{\varepsilon} \right] \\ & \quad \left. + \psi(z-\varepsilon)f_0(z-\varepsilon) - \psi(z+\varepsilon)g_0(z+\varepsilon) \right\} = 0. \end{aligned} \quad (2.4)$$

Furthermore,

$$g(w) = \frac{3}{4} \left[\int_0^1 u^{-\frac{1}{2}} (1-u)^{-\frac{1}{2}} (t-z+(w-t)u)^{\frac{3}{2}} du \right]_{t=w} = \frac{3\pi}{4} (w-z)^{\frac{3}{2}}. \quad (2.5)$$

Moreover,

$$f(t) = \frac{3}{4} (f_1(t) - 6f_2(t) + f_3(t)),$$

where

$$\begin{aligned} f_1(t) &= \int_s^w (w-s)^{-\frac{1}{2}} (s-z)^{\frac{3}{2}} (s-t)^{-\frac{1}{2}} ds \\ &= \int_0^{\frac{w-z}{w-t}} u^{-\frac{1}{2}} (1-u)^{-\frac{1}{2}} (w-z-(w-t)u)^{\frac{3}{2}} du, \\ f_2(t) &= \int_s^w (w-s)^{\frac{1}{2}} (s-z)^{\frac{1}{2}} (s-t)^{-\frac{1}{2}} ds \\ &= (z-t) \int_0^{\frac{w-z}{z-t}} u^{\frac{1}{2}} (1+u)^{-\frac{1}{2}} (w-z-(z-t)u)^{\frac{1}{2}} du, \\ f_3(t) &= \int_s^w (w-s)^{\frac{3}{2}} (s-z)^{-\frac{1}{2}} (s-t)^{-\frac{1}{2}} ds \\ &= \int_0^{\frac{w-z}{z-t}} u^{-\frac{1}{2}} (1+u)^{-\frac{1}{2}} (w-z-(z-t)u)^{\frac{3}{2}} du. \end{aligned}$$

It follows that

$$\begin{aligned} f'_1(t) &= \frac{3}{2} \int_0^{\frac{w-z}{w-t}} u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} (w-z-(w-t)u)^{\frac{1}{2}} du, \\ f'_2(t) &= - \int_0^{\frac{w-z}{z-t}} u^{\frac{1}{2}} (1+u)^{-\frac{1}{2}} (w-z-(z-t)u)^{\frac{1}{2}} du \\ & \quad + \frac{1}{2} (z-t) \int_0^{\frac{w-z}{z-t}} u^{\frac{3}{2}} (1+u)^{-\frac{1}{2}} (w-z-(z-t)u)^{-\frac{1}{2}} du \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{z-t} \left\{ - \int_s^w (w-s)^{\frac{1}{2}} (s-t)^{-\frac{1}{2}} [(s-z)^{\frac{1}{2}} - (s-t)^{\frac{1}{2}}] ds - \int_s^w (w-s)^{\frac{1}{2}} ds \right. \\
&\quad + \frac{1}{2} \int_s^w (w-s)^{-\frac{1}{2}} (s-t)^{-\frac{1}{2}} [(s-z)^{\frac{3}{2}} - (s-t)^{\frac{3}{2}}] ds \\
&\quad \left. + \frac{1}{2} \int_s^w (w-s)^{-\frac{1}{2}} (s-t) ds \right\} \\
&= \frac{1}{2} (w-t) \int_0^{\frac{w-z}{w-t}} u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \int_0^1 ((w-t)(1-u) + \theta(t-z))^{-\frac{1}{2}} d\theta \\
&\quad - \frac{3}{4} \int_0^{\frac{w-z}{w-t}} u^{-\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \int_0^1 ((w-t)(1-u) + \theta(t-z))^{\frac{1}{2}} d\theta + (w-z)^{\frac{1}{2}}, \\
f'_3(t) &= \frac{3}{2} \int_0^{\frac{w-z}{w-t}} u^{\frac{1}{2}} (1+u)^{-\frac{1}{2}} (w-z-(z-t)u)^{\frac{1}{2}} du \\
&= \frac{(w-z)^{\frac{3}{2}}}{z-t} - \frac{3}{4} (w-t) \int_0^{\frac{w-z}{w-t}} u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \int_0^1 ((w-t)(1-u) + \theta(t-z))^{-\frac{1}{2}} d\theta.
\end{aligned}$$

Thus

$$\begin{aligned}
f'(t) &= \frac{3}{4} \left\{ \frac{(w-z)^{\frac{3}{2}}}{z-t} + \frac{3}{2} \int_0^{\frac{w-z}{w-t}} u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} (w-z-(w-t)u)^{\frac{1}{2}} du \right. \\
&\quad - \frac{15}{4} (w-t) \int_0^{\frac{w-z}{w-t}} u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \int_0^1 ((w-t)(1-u) + \theta(t-z))^{-\frac{1}{2}} d\theta \\
&\quad + \frac{9}{2} \int_0^{\frac{w-z}{w-t}} u^{-\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \int_0^1 ((w-t)(1-u) + \theta(t-z))^{\frac{1}{2}} d\theta \\
&\quad \left. - 6(w-z)^{\frac{1}{2}} \right\}.
\end{aligned}$$

Similarly, we may obtain

$$\begin{aligned}
g'(t) &= \frac{3}{4} \left\{ \frac{(w-z)^{\frac{3}{2}}}{z-t} + \frac{3}{2} \int_0^1 u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} (w-z-(w-t)u)^{\frac{1}{2}} du \right. \\
&\quad - \frac{15}{4} (w-t) \int_0^1 u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \int_0^1 ((w-t)(1-u) + \theta(t-z))^{-\frac{1}{2}} d\theta \\
&\quad + \frac{9}{2} \int_0^1 u^{-\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \int_0^1 ((w-t)(1-u) + \theta(t-z))^{\frac{1}{2}} d\theta \\
&\quad \left. + \frac{3}{2} \int_0^1 (w-z+\theta(z-t))^{\frac{1}{2}} d\theta \right\}.
\end{aligned}$$

Therefore,

$$\begin{aligned}
&\lim_{\varepsilon \rightarrow 0} \left\{ \int_{-1}^{s-\varepsilon} \psi(t) f'(t) dt + \int_{s+\varepsilon}^w \psi(t) g'(t) dt \right\} \\
&= \frac{3}{4} \left\{ (w-z)^{\frac{3}{2}} \int_{-1}^w \frac{\psi(t)}{z-t} dt + F(w, z) \right\}. \quad (2.6)
\end{aligned}$$

Here

$$F(w, z) = \frac{3}{2} F_1(w, z) - \frac{15}{4} F_2(w, z),$$

where

$$\begin{aligned} F_1(w, z) = & \int_{-1}^w \psi(t) dt \int_0^{\frac{w-\chi(z,t)}{w-t}} u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} (w-z-(w-t)u)^{\frac{1}{2}} du \\ & + 3 \int_{-1}^w \psi(t) dt \int_0^{\frac{w-\chi(z,t)}{w-t}} u^{-\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \\ & \times \int_0^1 ((w-t)(1-u) + \theta(t-z))^{\frac{1}{2}} d\theta \\ & - 4(w-z)^{\frac{1}{2}} \int_{-1}^z \psi(t) dt + \int_z^w \psi(t) dt \int_0^1 (w-z+\theta(z-t))^{\frac{1}{2}} d\theta, \end{aligned}$$

$$\begin{aligned} F_2(w, z) = & \int_{-1}^w (w-t) \psi(t) dt \int_0^{\frac{w-\chi(z,t)}{w-t}} u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \\ & \times \int_0^1 ((w-t)(1-u) + \theta(t-z))^{-\frac{1}{2}} d\theta, \end{aligned}$$

$$\chi(z, t) = \begin{cases} z, & -1 < t \leq z, \\ t, & z < t \leq w. \end{cases}$$

From (2.3)–(2.6), we have

$$\eta(w, z) = \frac{3}{4} \left\{ (w-z)^{\frac{3}{2}} \left(\pi \psi(w) - \int_{-1}^w \frac{\psi(t)}{z-t} dt \right) - F(w, z) \right\}. \quad (2.7)$$

Let $\psi \in C_0^\infty(-1, 1)$ satisfy $\text{supp } \psi \subset [-1+s, 1-s]$ ($s < \frac{1}{6}$) and

$$\int_{-1}^1 \psi(t) dt = 0.$$

Set $\psi_n(t) = n\psi(nt)$ and replace $\psi(t)$ with $\psi_n(t)$ in (2.7), then

$$\eta_n(w, z) = \frac{3}{4} \left\{ (w-z)^{\frac{3}{2}} \left(\pi \psi_n(w) - \int_{-1}^w \frac{\psi_n(t)}{z-t} dt \right) - F_n(w, z) \right\}, \quad (2.8)$$

$$\begin{aligned} \tau_n(w, z) = & -\frac{15}{32} \left\{ -\frac{2}{5} (w-z)^{\frac{5}{2}} \int_{-1}^w \frac{\psi_n(t)}{z-t} dt \right. \\ & + \left(\pi - \frac{2}{5} \right) \int_z^w (t-z)^{\frac{3}{2}} \psi_n(t) dt - \int_z^w F_n(x, z) dx \Big\} \\ = & -\frac{15}{32} \left\{ \pi (w-z)^{\frac{3}{2}} \int_z^w \psi_n(t) dt + \pi \int_z^w [(t-z)^{\frac{3}{2}} - (w-z)^{\frac{3}{2}}] \psi_n(t) dt \right. \\ & - \frac{2}{5} \int_{-1}^z (z-t)^{\frac{3}{2}} \psi_n(t) dt - \frac{2}{5} \int_{-1}^w \frac{\psi_n(t)}{z-t} [(w-z)^{\frac{5}{2}} - |t-z|^{\frac{5}{2}}] dt \\ & \left. - \int_z^w F_n(x, z) dx \right\}, \quad (2.9) \end{aligned}$$

$$\bar{\tau}_n(w, z) = \frac{15}{32} \left\{ \frac{2}{5} \pi (w-z)^{\frac{5}{2}} \psi_n(w) - \int_s^w (w-y)^{\frac{3}{2}} dy \int_{-1}^w \frac{\psi_n(t)}{y-t} dt \right. \\ \left. - \int_s^w F_n(w, y) dy \right\}, \quad (2.10)$$

where

$$F_n(w, z) = \frac{3}{2} F_1^n(w, z) - \frac{15}{4} F_2^n(w, z), \quad (2.11)$$

while

$$\left. \begin{aligned} F_1^n(w, z) &= \int_{-1}^w \psi_n(t) dt \int_0^{\frac{w-\chi(z,t)}{w-t}} u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} (w-z-(w-t)u)^{\frac{1}{2}} du \\ &\quad + 3 \int_{-1}^w \psi_n(t) dt \int_0^{\frac{w-\chi(z,t)}{w-t}} u^{-\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \int_0^1 ((w-t)(1-u) \\ &\quad + \theta(t-z))^{\frac{1}{2}} d\theta - 4(w-z)^{\frac{1}{2}} \int_{-1}^z \psi_n(t) dt \\ &\quad + \int_s^w \psi_n(t) dt \int_0^1 (w-z+\theta(z-t))^{\frac{1}{2}} d\theta, \\ F_2^n(w, z) &= \int_{-1}^w (w-t) \psi_n(t) dt \int_0^{\frac{w-\chi(z,t)}{w-t}} u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \\ &\quad \times \int_0^1 ((w-t)(1-u) + \theta(t-z))^{-\frac{1}{2}} d\theta. \end{aligned} \right\} \quad (2.12)$$

We obtain by computation

$$\left. \begin{aligned} \int_s^w F_1^n(x, z) dx &= \frac{2}{3} \left\{ \int_{-1}^w \psi_n(t) dt \int_0^{\frac{w-\chi(z,t)}{w-t}} u^{\frac{1}{2}} (1-u)^{-\frac{3}{2}} [(w-z-(w-t)u)^{\frac{3}{2}} \right. \\ &\quad \left. - (\chi(z, t) - z)^{\frac{3}{2}}] du - 4(w-z)^{\frac{3}{2}} \int_{-1}^z \psi_n(t) dt \right. \\ &\quad + 9 \int_{-1}^w (w-t) \psi_n(t) dt \int_0^{\frac{w-\chi(z,t)}{w-t}} u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \\ &\quad \times \int_0^1 ((w-t)(1-u) + \theta(t-z))^{\frac{1}{2}} d\theta \\ &\quad \left. + \int_s^w \psi_n(t) dt \int_0^1 (w-z+\theta(z-t))^{\frac{3}{2}} - (t-z)^{\frac{3}{2}} (1-\theta)^{\frac{3}{2}} d\theta \right\}, \\ \int_s^w F_2^n(x, z) dx &= \frac{2}{3} \int_{-1}^w (w-t)^2 \psi_n(t) dt \int_0^{\frac{w-\chi(z,t)}{w-t}} u^{\frac{3}{2}} (1-u)^{-\frac{1}{2}} du \\ &\quad \times \int_0^1 ((w-t)(1-u) + \theta(t-z))^{-\frac{1}{2}} d\theta, \end{aligned} \right\} \quad (2.13)$$

and

$$\begin{aligned}
\int_0^w F_1^n(w, y) dy = & \frac{2}{3} \left\{ \int_{-1}^w \psi_n(t) dt \int_0^{\frac{w-X(z,t)}{w-t}} u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} (w-z-(w-t)u)^{\frac{3}{2}} du \right. \\
& + \int_0^w (t-z) \psi_n(t) dt \int_0^1 d\theta \\
& \times \int_0^1 [w-z-(\theta+(1-\theta)\tau)(t-z)]^{\frac{1}{2}} d\tau \\
& + 3 \int_{-1}^w \psi_n(t) dt \int_0^{\frac{w-X(z,t)}{w-t}} u^{-\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \\
& \times \int_0^1 \frac{((w-t)(1-u)+\theta(t-z))^{\frac{3}{2}} - ((1-\theta)(1-u)(w-t))^{\frac{3}{2}}}{\theta} d\theta, \\
& \left. - 4 \int_{-1}^w (w-X(z,t))^{\frac{3}{2}} \psi_n(t) dt \right\}, \\
\int_0^w F_2^n(w, y) dy = & 2 \int_{-1}^w (w-t) \psi_n(t) dt \int_0^{\frac{w-X(z,t)}{w-t}} u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \\
& \times \int_0^1 \frac{((w-t)(1-u)+\theta(t-z))^{\frac{1}{2}} - ((1-\theta)(1-u)(w-t))^{\frac{1}{2}}}{\theta} d\theta.
\end{aligned} \tag{2.14}$$

O. For (η_0, q_0) , (η_1, q_1) and (η_n, q_n) gained in (A) and (B), we have the following results on any bounded domain $K \subset \{(w, z): w \geq z\}$.

Lemma 1.

$$|B_n^0| = |\eta_0 q_n - \eta_n q_0| \leq C n^{-\frac{1}{4}},$$

and

$$|B_n^1| = |\eta_1 q_n - \eta_n q_1| \leq C n^{-\frac{1}{4}}$$

Proof. Above all, we observe that the following inequality

$$|x^\alpha - y^\alpha| \leq |x - y|^\alpha, \quad \text{for } 0 < \alpha \leq 1 \tag{2.15}$$

holds.

From the expression of (η_0, q_0) and (η_n, q_n) , we obtain

$$\begin{aligned}
B_n^0 = & -\frac{15}{32} (-zw)^{\frac{3}{2}} X \left\{ -\frac{2}{5} (w-z)^{\frac{3}{2}} \left(\pi w \psi_n(w) + z \int_{-1}^w \frac{\psi_n(t)}{z-t} dt \right) \right. \\
& \left. + \left(\pi - \frac{2}{5} \right) \int_0^w (t-z)^{\frac{3}{2}} \psi_n(t) dt + \frac{2w}{5} F_n(w, z) - \int_0^w F_n(x, z) dx \right\}.
\end{aligned} \tag{2.16}$$

Remark that B_n^0 vanish on $G = \{(w, z): w > -\frac{1}{n}, z < \frac{1}{n}\}^c$, while

$$G^c \cap K = Q \cup I \cup \bar{S}_w \cup \bar{S}_n,$$

where

$$Q = \left\{ (w, z): w \geq \frac{1}{n}, z \leq -\frac{1}{n} \right\} \cap K,$$

$$I = \left\{ (w, z) : |w| \leq \frac{1}{n}, \quad |z| \leq \frac{1}{n} \right\} \cap G^0 \cap K,$$

$$\tilde{S}_w = \left\{ (w, z) : |w| \leq \frac{1}{n}, \quad z < -\frac{1}{n} \right\} \cap K,$$

$$\tilde{S}_z = \left\{ (w, z) : w > \frac{1}{n}, \quad |z| \leq \frac{1}{n} \right\} \cap K.$$

On the wake quadrant Q , $\psi_n(w) = 0$ and

$$B_n^0 = -\frac{15}{32} (-zw)^{\frac{3}{2}} \left\{ -\frac{2}{5} (w-z)^{\frac{3}{2}} \int_{-1}^1 \frac{\psi(t)}{1 - \frac{t}{nz}} dt \right. \\ \left. + \left(\pi - \frac{2}{5} \right) \int_{-1}^1 \left(\frac{t}{n} - z \right)^{\frac{3}{2}} \psi(t) dt + \frac{2}{5} w F_n(w, z) - \int_z^w F_n(x, z) dx \right\}.$$

Using the inequality (2.14) and $\int_{-1}^1 \psi(t) dt = 0$ we have

$$|F_1^n(w, z)| = \left| \int_{-1}^1 \psi(t) dt \left\{ \int_0^1 u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} \left[\left(w-z - \left(w - \frac{t}{n} \right) u \right)^{\frac{1}{2}} \right. \right. \right. \right. \\ \left. \left. \left. - (w-z-wu)^{\frac{1}{2}} \right] du + 3 \int_0^1 u^{-\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \right. \right. \\ \left. \left. \times \int_0^1 \left[\left(\left(w - \frac{t}{n} \right) (1-u) + \theta \left(\frac{t}{n} - z \right) \right)^{\frac{1}{2}} - (w(1-u) - \theta z)^{\frac{1}{2}} \right] d\theta \right\} \right. \\ \left. + \int_{-1}^1 \psi(t) dt \int_0^1 \left[\left(w-z + \theta \left(z - \frac{t}{n} \right) \right)^{\frac{1}{2}} - (w-z + \theta z)^{\frac{1}{2}} \right] d\theta \right| \\ \leq \frac{1}{\sqrt{n}} \left\{ \int_{-1}^1 |\psi(t)| dt \int_0^1 u(1-u)^{-\frac{1}{2}} du + 3 \int_{-1}^1 |\psi(t)| dt \right. \\ \left. \times \int_0^1 u^{-\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \int_0^1 |\theta + u - 1|^{\frac{1}{2}} d\theta + \int_{-1}^1 |\psi(t)| dt \int_0^1 \theta^{\frac{1}{2}} d\theta \right\} \\ \leq \frac{C}{\sqrt{n}}, \\ |F_2^n(w, z)| = \left| \int_{-1}^1 \psi(t) dt \int_0^1 u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \right. \\ \left. \times \int_0^1 \left[\left(w - \frac{t}{n} \right) \left(\left(w - \frac{t}{n} \right) (1-u) + \theta \left(\frac{t}{n} - z \right) \right)^{-\frac{1}{2}} \right. \right. \\ \left. \left. - w(w(1-u) - \theta z)^{-\frac{1}{2}} \right] d\theta \right| \\ \leq \int_{-1}^1 |\psi(t)| dt \int_0^1 u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \int_0^1 \left| \frac{-\frac{t}{n}}{\left(\left(w - \frac{t}{n} \right) (1-u) + \theta \left(\frac{t}{n} - z \right) \right)^{\frac{1}{2}}} \right| d\theta$$

$$\begin{aligned}
& + w \frac{\left(\left(w - \frac{t}{n} \right) (1-u) + \theta \left(\frac{t}{n} - z \right) \right)^{\frac{1}{2}} - (w(1-u) - \theta z)^{\frac{1}{2}}}{\left(\left(w - \frac{t}{n} \right) (1-u) + \theta \left(\frac{t}{n} - z \right) \right)^{\frac{1}{2}} (w(1-u) - \theta z)^{\frac{1}{2}}} d\theta \\
& \leq \frac{1}{\sqrt{n}} \int_{-1+s}^{1-s} \frac{|\psi(t)|}{\sqrt{1-\frac{t}{nz}}} dt \int_0^1 u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \int_0^1 \theta^{-\frac{1}{2}} d\theta \\
& \quad + n^{\frac{1}{4}} (-nz)^{-\frac{3}{4}} w \int_{-1+s}^{1-s} \frac{|\psi(t)|}{\sqrt{1-\frac{t}{nz}}} dt \int_0^1 u^{\frac{1}{2}} (1-u)^{-\frac{3}{4}} du \int_0^1 |u-\theta|^{\frac{1}{2}} \theta^{-\frac{3}{4}} d\theta \\
& \leq O(n^{-\frac{1}{2}} + n^{\frac{1}{4}} (-nz)^{-\frac{3}{4}} w).
\end{aligned}$$

Hence

$$|F_n(w, z)| \leq O(n^{-\frac{1}{2}} + n^{\frac{1}{4}} (-nz)^{-\frac{3}{4}} w). \quad (2.17)$$

Similarly, we get

$$\left| \int_s^w F_n(x, z) dx \right| \leq O(n^{-\frac{1}{2}} + n^{-\frac{1}{4}} (-z)^{-\frac{1}{2}} w).$$

Furthermore, we have

$$\begin{aligned}
& \left| (-z)^{\frac{1}{2}} \int_{-1}^1 \frac{\psi(t)}{1-\frac{t}{nz}} dt \right| = \left| \int_{-1}^1 \frac{\sqrt{-z}}{nz} \frac{t\psi(t)}{1-\frac{t}{nz}} dt \right| \leq \frac{C}{\sqrt{n}}, \\
& \left| \int_{-1}^1 \left(\frac{t}{n} - z \right)^{\frac{3}{2}} \psi(t) dt \right| = \left| \int_{-1}^1 \left[\left(\frac{t}{n} - z \right)^{\frac{3}{2}} - (-z)^{\frac{3}{2}} \right] \psi(t) dt \right| \\
& \leq \frac{3}{2n} \left| \int_{-1}^1 \psi(t) dt \int_0^1 \left(-z + \theta \frac{t}{n} \right)^{\frac{1}{2}} d\theta \right| \leq \frac{C}{n}.
\end{aligned}$$

So

$$|B_n^0| \leq Cn^{-\frac{1}{4}}. \quad (2.18)$$

On the interaction zone I,

$$\begin{aligned}
B_n^0 = & -\frac{15}{32} (-zw)^{\frac{3}{2}} X \left\{ -\frac{2}{5} (w-z)^{\frac{3}{2}} \left(\pi n w \psi(nw) + nz \int_{-1}^{nw} \frac{\psi(t)}{nz-t} dt \right) \right. \\
& \left. + \left(\pi - \frac{2}{5} \right) \int_{nw}^{nw} \left(\frac{t}{n} - z \right)^{\frac{3}{2}} \psi(t) dt + \frac{2}{5} w F_n(w, z) - \int_s^w F_n(x, z) dx \right\}.
\end{aligned}$$

Observe that

$$\begin{aligned}
|F_n^1(w, z)| = & \left| \int_{-1}^{nw} \psi(t) dt \int_0^{\frac{w-z(s, \frac{t}{n})}{w-\frac{t}{n}}} u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} \left(w-z - \left(w - \frac{t}{n} \right) u \right)^{\frac{1}{2}} du \right. \\
& \left. + 3 \int_{-1}^{nw} \psi(t) dt \int_0^{\frac{w-z(s, \frac{t}{n})}{w-\frac{t}{n}}} u^{-\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \right|
\end{aligned}$$

$$\begin{aligned}
& \times \int_0^1 \left(\left(w - \frac{t}{n} \right) (1-u) + \theta \left(\frac{t}{n} - z \right) \right)^{\frac{1}{2}} d\theta - 4(w-z)^{\frac{1}{2}} \int_{-1}^{nw} \psi(t) dt \\
& + \int_{nz}^{nw} \psi(t) dt \int_0^1 \left(w - z + \theta \left(z - \frac{t}{n} \right) \right)^{\frac{1}{2}} d\theta \Big| \leq \frac{O}{\sqrt{n}}, \\
|F_2^n(w, z)| &= \left| \int_{-1}^{nw} \left(w - \frac{t}{n} \right) \psi(t) dt \int_0^{\frac{w-z(z, \frac{t}{n})}{w-\frac{t}{n}}} u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} \right. \\
& \quad \times \frac{\left(\left(w - \frac{t}{n} \right) (1-u) + \left(\frac{t}{n} - z \right) \right)^{\frac{1}{2}} - \left(\left(w - \frac{t}{n} \right) (1-u) \right)^{\frac{1}{2}}}{\frac{t}{n} - z} du \Big| \\
& \leq B\left(\frac{1}{2}, \frac{3}{2}\right) \int_{-1}^{nw} \frac{w - \frac{t}{n}}{\sqrt{\frac{t}{n} - z}} |\psi(t)| dt \leq \frac{O}{\sqrt{n}},
\end{aligned}$$

then

$$\begin{aligned}
|F_n(w, z)| &\leq \frac{O}{\sqrt{n}}, \\
\left| \int_z^w F_n(x, z) dx \right| &\leq \frac{O(w-z)}{\sqrt{n}} \leq O n^{-\frac{3}{2}}.
\end{aligned}$$

Also,

$$\left| (w-z)^{\frac{3}{2}} \left(\pi n w \psi(nw) + n z \int_{-1}^{nw} \frac{\psi(t)}{nz-t} dt \right) \right| + \left| \int_{nz}^{nw} \left(\frac{t}{n} - z \right)^{\frac{3}{2}} \psi(t) dt \right| \leq O n^{-\frac{3}{2}}.$$

It follows that

$$|\eta_n| + |\tau_n| \leq \frac{O}{\sqrt{n}}, \quad (2.19)$$

and

$$|B_n^0| \leq O n^{-\frac{3}{2}}. \quad (2.20)$$

On the truncated strips $\tilde{S}_w$,

$$\begin{aligned}
B_n^0 &= -\frac{15}{32} (-zw)^{\frac{3}{2}} X \left\{ -\frac{2}{5} (w-z)^{\frac{3}{2}} \left(\pi n w \psi(nw) + \int_{-1}^{nw} \frac{\psi(t)}{1 - \frac{t}{nz}} dt \right) \right. \\
& \quad + \left(\pi - \frac{2}{5} \right) \int_{-1}^{nw} \left(\frac{t}{n} - z \right)^{\frac{3}{2}} \psi(t) dt \\
& \quad \left. + \frac{2}{5} w F_n(w, z) - \int_z^w F_n(x, z) dx \right\}.
\end{aligned}$$

In this case,

$$\begin{aligned}
|F_1^n(w, z)| &= \left| \int_{-1}^{nw} \psi(t) dt \int_0^1 u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} \left(w-z - \left(w-\frac{t}{n} \right) u \right)^{\frac{1}{2}} du \right. \\
&\quad + 3 \int_{-1}^{nw} \psi(t) dt \int_0^1 u^{-\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \\
&\quad \times \int_0^1 \left(\left(w-\frac{t}{n} \right) (1-u) + \theta \left(\frac{t}{n} - z \right) \right)^{\frac{1}{2}} d\theta \\
&\quad \left. + \int_{ns}^{nw} \psi(t) dt \int_0^1 \left(w-z + \theta \left(z-\frac{t}{n} \right) \right)^{\frac{1}{2}} d\theta \right| \leq O, \\
|F_2^n(w, z)| &= \left| \int_{-1}^{nw} \left(w-\frac{t}{n} \right) \psi(t) dt \int_0^1 u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \right. \\
&\quad \times \int_0^1 \left(\left(w-\frac{t}{n} \right) (1-u) + \theta \left(\frac{t}{n} - z \right) \right)^{-\frac{1}{2}} d\theta \Big| \\
&\leq B \left(\frac{1}{2}, \frac{3}{2} \right) \int_0^1 \theta^{-\frac{1}{2}} d\theta \cdot \frac{1}{\sqrt{n}} \left| \int_{-1}^{nw} \frac{nw-t}{\sqrt{t-nz}} \psi(t) dt \right| \leq \frac{O}{\sqrt{n}}.
\end{aligned}$$

So

$$|F_n(w, z)| \leq O. \quad (2.21)$$

Also,

$$\begin{aligned}
\left| \int_s^w F_1^n(x, z) dx \right| &= \frac{2}{3} \left| \frac{3}{2} \int_{-1}^{nw} \psi(t) dt \int_0^1 u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \right. \\
&\quad \times \int_0^1 \left(\frac{t}{n} - z + \theta(1-u) \left(w-\frac{t}{n} \right) \right)^{\frac{1}{2}} \left(w-\frac{t}{n} \right) d\theta \\
&\quad + 4 \int_{-1}^{nw} \left(w-\frac{t}{n} \right) \psi(t) dt \int_0^1 u^{-\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \\
&\quad \times \int_0^1 \left(\left(w-\frac{t}{n} \right) (1-u) + \theta \left(\frac{t}{n} - z \right) \right)^{\frac{1}{2}} d\theta + \frac{3}{2} \int_{-1}^{nw} \psi(t) dt \\
&\quad \times \int_0^1 d\theta \int_0^1 \left(\left(\frac{t}{n} - z \right) (1-\theta) + \left(w-\frac{t}{n} \right) \tau \right)^{\frac{1}{2}} \left(w-\frac{t}{n} \right) d\tau \Big| \\
&\leq \frac{O}{n},
\end{aligned}$$

$$\begin{aligned}
\left| \int_s^w F_2^n(x, z) dx \right| &= \frac{2}{3} \left| \int_{-1}^{nw} \left(w-\frac{t}{n} \right)^2 \psi(t) dt \int_0^1 u^{\frac{3}{2}} (1-u)^{-\frac{1}{2}} du \right. \\
&\quad \times \int_0^1 \left(\left(w-\frac{t}{n} \right) (1-u) + \theta \left(\frac{t}{n} - z \right) \right)^{-\frac{1}{2}} d\theta \Big| \\
&\leq O \left| \int_{-1}^{nw} \frac{\left(w-\frac{t}{n} \right)^2}{\sqrt{\frac{t}{n} - z}} \psi(t) dt \right| \\
&\leq O n^{-\frac{3}{2}} \int_{-1}^{nw} \frac{(nw-t)^2}{\sqrt{t-nz}} |\psi(t)| dt \leq O n^{-\frac{3}{2}}.
\end{aligned}$$

Hence

$$\left| \int_s^w F_n(x, z) dx \right| \leq \frac{C}{n}. \quad (2.22)$$

From (2.21), (2.22) and

$$\left| \pi n w \psi(nw) + \int_{-1}^{nw} \frac{\psi(t)}{1 - \frac{t}{nz}} dt \right| \leq \pi |\psi(nw)| + \int_{-1+s}^{\min(nw, 1-s)} \frac{|\psi(t)|}{s} dt \leq O,$$

we immediately obtain

$$|B_n^0| \leq O n^{-\frac{3}{2}}. \quad (2.23)$$

On the other truncated strips $\bar{S}_s$:

$$\begin{aligned} B_n^0 = & -\frac{15}{32} (-zw)^{\frac{3}{2}} X \left\{ -\frac{2}{5} (w-z)^{\frac{3}{2}} nz \int_{-1}^1 \frac{\psi(t)}{nz-t} dt \right. \\ & \left. + \left(\pi - \frac{2}{5} \right) \int_{nz}^1 \left(\frac{t}{n} - z \right)^{\frac{3}{2}} \psi(t) dt + \frac{2}{5} w F_n(w, z) - \int_s^w F_n(x, z) dx \right\}. \end{aligned}$$

Since

$$\begin{aligned} |F_1^n(w, z)| = & \left| \int_{-1}^1 \psi(t) dt \int_0^{\frac{w-\chi(s, \frac{t}{n})}{w-\frac{t}{n}}} u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} \left(w-z - \left(w - \frac{t}{n} \right) u \right)^{\frac{1}{2}} du \right. \\ & + \int_{nz}^1 \psi(t) dt \int_0^1 \left(w-z + \theta \left(z - \frac{t}{n} \right) \right)^{\frac{1}{2}} d\theta \\ & + 3 \int_{-1}^1 \psi(t) dt \int_0^{\frac{w-\chi(s, \frac{t}{n})}{w-\frac{t}{n}}} u^{-\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \\ & \times \int_0^1 \left(\left(w - \frac{t}{n} \right) (1-u) + \theta \left(\frac{t}{n} - z \right) \right)^{\frac{1}{2}} d\theta - 4(w-z)^{\frac{1}{2}} \int_{-1}^{nz} \psi(t) dt \Big| \\ \leq & B\left(\frac{1}{2}, \frac{3}{2}\right) (w-z)^{\frac{1}{2}} \int_{-1}^1 |\psi(t)| dt + \left(w + \frac{1}{n} \right)^{\frac{1}{2}} \\ & \times \left| \int_{nz}^1 |\psi(t)| dt + 4(w-z)^{\frac{1}{2}} \left| \int_{-1}^{nz} \psi(t) dt \right| \right| \\ & + 3 \left(w + \frac{1}{n} \right)^{\frac{1}{2}} \int_{-1}^1 |\psi(t)| dt \int_0^1 u^{-\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \\ & \times \int_0^1 \left(1-u + \theta \frac{t-nz}{nw-t} \right)^{\frac{1}{2}} d\theta \leq O \left(w + \frac{1}{n} \right)^{\frac{1}{2}}, \\ |F_2^n(w, z)| = & \left| \int_{-1}^1 \psi(t) dt \int_{\chi(z, \frac{t}{n})}^w (w-s)^{\frac{1}{2}} \left(s - \frac{t}{n} \right)^{-\frac{1}{2}} ds \right. \\ & \times \left. \int_0^1 \left(s - \frac{t}{n} + \theta \left(\frac{t}{n} - z \right) \right)^{-\frac{1}{2}} d\theta \right| \end{aligned}$$

$$\begin{aligned}
& - \left| \int_{-1}^1 \psi(t) dt \int_{x(s, \frac{t}{n})}^w \frac{(w-s)^{\frac{1}{2}} \left(s - \frac{t}{n}\right)^{-\frac{1}{2}}}{\sqrt{s-z} + \sqrt{s - \frac{t}{n}}} ds \right| \\
& \leq \int_{-1}^{ns} |\psi(t)| dt \int_s^w (w-s)^{\frac{1}{2}} (s-z)^{-\frac{1}{2}} \left(s - \frac{t}{n}\right)^{-\frac{1}{2}} ds \\
& \quad + \int_{ns}^1 |\psi(t)| dt \int_{\frac{t}{n}}^w (w-s)^{\frac{1}{2}} (s-z)^{-\frac{1}{2}} \left(s - \frac{t}{n}\right)^{-\frac{1}{2}} ds \\
& \leq \begin{cases} \int_{-1}^{ns} |\psi(t)| \left(z - \frac{t}{n}\right)^{-\frac{1}{2}} dt \int_s^w (w-s)^{\frac{1}{2}} (s-z)^{-\frac{1}{2}} ds \\
\quad + \int_{ns}^1 |\psi(t)| \left(\frac{t}{n} - z\right)^{-\frac{1}{2}} dt \int_{\frac{t}{n}}^w (w-s)^{\frac{1}{2}} \left(s - \frac{t}{n}\right)^{-\frac{1}{2}} ds \\
\text{or} \\
\int_{-1}^{ns} |\psi(t)| dt \int_s^w (w-s)^{\frac{1}{2}} (s-z)^{-\frac{3}{4}} \left(s - \frac{t}{n}\right)^{-\frac{1}{4}} ds \\
\quad + \int_{ns}^1 |\psi(t)| dt \int_{\frac{t}{n}}^w (w-s)^{\frac{1}{2}} (s-z)^{-\frac{1}{4}} \left(s - \frac{t}{n}\right)^{-\frac{3}{4}} ds \\
B\left(\frac{1}{2}, \frac{3}{2}\right) \left[(w-z) \int_{-1}^{ns} \frac{|\psi(t)|}{\sqrt{z - \frac{t}{n}}} dt + \int_{ns}^1 \frac{|\psi(t)|}{\sqrt{\frac{t}{n} - z}} \left(w - \frac{t}{n}\right) dt \right] \\
\leq O\left(w + \frac{1}{n}\right) \int_{-1}^1 \frac{|\psi(t)|}{\left|z - \frac{t}{n}\right|^{\frac{1}{4}}} dt \leq O n^{\frac{1}{2}} \left(w + \frac{1}{n}\right) \\
\text{or} \\
B\left(\frac{3}{2}, \frac{1}{4}\right) \left[(w-z)^{\frac{3}{4}} \int_{-1}^{ns} \frac{|\psi(t)|}{\left(z - \frac{t}{n}\right)^{\frac{1}{4}}} dt + \int_{ns}^1 \frac{|\psi(t)|}{\left(\frac{t}{n} - z\right)^{\frac{1}{4}}} \left(w - \frac{t}{n}\right)^{\frac{3}{4}} dt \right] \\
\leq O\left(w + \frac{1}{n}\right)^{\frac{3}{4}} \int_{-1}^1 \frac{|\psi(t)|}{\left|z - \frac{t}{n}\right|^{\frac{1}{4}}} dt \leq O n^{\frac{1}{4}} \left(w + \frac{1}{n}\right)^{\frac{3}{4}}
\end{cases}
\end{aligned}$$

$$\leq O \min \left(n^{\frac{1}{2}} ((w-z) + n^{-1}), n^{\frac{1}{4}} ((w-z)^{\frac{3}{4}} + n^{-\frac{3}{4}}) \right),$$

we have

$$|F_n(w, z)| \leq O \min \left(n^{\frac{1}{2}} ((w-z) + n^{-1}), n^{\frac{1}{4}} ((w-z)^{\frac{3}{4}} + n^{-\frac{3}{4}}) \right), \quad (2.24)$$

$$\left| \int_s^w F_n(x, z) dx \right| \leq O n^{\frac{1}{4}} (w-z) ((w-z)^{\frac{3}{4}} + n^{-\frac{3}{4}}) \leq O n^{\frac{1}{4}}.$$

Also,

$$\left|nz \int_{-1}^1 \frac{\psi(t)}{nz-t} dt\right| \leq \left|\int_{-1}^1 \frac{\psi(t)}{nz-t} dt\right| \leq O,$$

$$\left|\int_{ns}^1 \left(\frac{t}{n}-z\right)^{\frac{3}{2}} \psi(t) dt\right| \leq n^{-\frac{3}{2}} \left|\int_{ns}^1 (t-nz)^{\frac{3}{2}} \psi(t) dt\right| \leq On^{-\frac{3}{2}}.$$

Thus,

$$|B_n^0| \leq On^{-\frac{5}{4}}, \quad (2.25)$$

It follows from (2.18), (2.20), (2.23) and (2.25) that

$$|B_n^0| \leq On^{-\frac{1}{4}}.$$

In the same fashion, we obtain

$$|B_n^1| = |\eta_1 q_n - \eta_n q_1| \leq On^{-\frac{1}{4}}.$$

This completes the proof of Lemma 1. ■

D. For the above $\psi(t)$, we define

$$\hat{\psi}(t) = \int_{-1}^t \psi(s) ds + t\psi(t),$$

then $\hat{\psi}(t) \in C_0^\infty(-1, 1)$, $\text{supp } \hat{\psi} \subset (-1+\varepsilon, 1-\varepsilon)$ and $\int_{-1}^1 \hat{\psi}(t) dt = 0$.

For the two weak-entropy pairs (η_n, q_n) and $(\hat{\eta}_n, \hat{q}_n)$ gained in the way in (B), we consider

$$B_n = \eta_n \hat{q}_n - \hat{\eta}_n q_n$$

on any bounded domain $K \subset \{(w, z): w \geq z\}$.

Recall that $G^0 \cap K = Q \cup I \cup \tilde{S}_w \cup \tilde{S}_z$, while

$$G \cap K = U_1 \cup U_2,$$

where

$$U_1 = \left\{ (w, z): w < -\frac{1}{n}, z < -\frac{1}{n} \right\} \cap K$$

and

$$U_2 = \left\{ (w, z): w > \frac{1}{n}, z > \frac{1}{n} \right\} \cap K.$$

On $\tilde{S}_w$,

$$\eta_n(w, z) = \frac{3}{4} \left\{ \pi n (w-z)^{\frac{3}{2}} \psi(nw) - (w-z)^{\frac{3}{2}} \int_{-1}^{nw} \frac{\psi(t)}{z - \frac{t}{n}} dt - F_n(w, z) \right\},$$

$$\tau_n(w, z) = -\frac{15}{92} \left\{ \pi (w-z)^{\frac{3}{2}} \int_{-1}^{nw} \psi(t) dt + \pi \int_{-1}^{nw} \left[\left(\frac{t}{n} - z \right)^{\frac{3}{2}} - (w-z)^{\frac{3}{2}} \right] \psi(t) dt \right. \\ \left. + \frac{2}{5} \int_{-1}^{nw} \frac{\psi(t)}{z - \frac{t}{n}} \left[\left(\frac{t}{n} - z \right)^{\frac{5}{2}} - (w-z)^{\frac{5}{2}} \right] dt - \int_{\cdot}^w F_n(x, z) dx \right\},$$

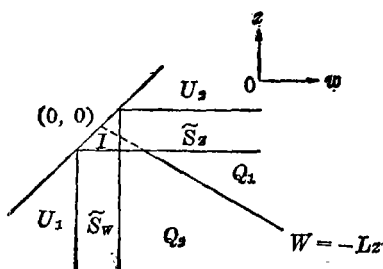


Fig. 1

then

$$B_n = \eta_n \hat{\tau}_n - \hat{\eta}_n \tau_n = -\frac{45}{128} \{ n(w-z)^3 A_1(nw) + (w-z)^{\frac{3}{2}} A_2(w, z) + A_3(w, z) \}, \quad (2.26)$$

where

$$\left. \begin{aligned} A_1(nw) &= \pi^2 \left[\psi(nw) \int_{-1}^{nw} \hat{\psi}(t) dt - \hat{\psi}(nw) \int_{-1}^{nw} \psi(t) dt \right] = \pi^2 \left(\int_{-1}^{nw} \psi(t) dt \right)^2, \\ A_2(w, z) &= n\pi\psi(nw) \left\{ \pi \int_{-1}^{nw} \left[\left(\frac{t}{n} - z \right)^{\frac{3}{2}} - (w-z)^{\frac{3}{2}} \right] \hat{\psi}(t) dt \right. \\ &\quad + \frac{2}{5} \int_{-1}^{nw} \left[\left(\frac{t}{n} - z \right)^{\frac{5}{2}} - (w-z)^{\frac{5}{2}} \right] \frac{\hat{\psi}(t)}{z - \frac{t}{n}} dt - \int_{\frac{z}{n}}^w \hat{F}_n(x, z) dx \Big\} \\ &\quad - n\pi\hat{\psi}(nw) \left\{ \pi \int_{-1}^{nw} \left[\left(\frac{t}{n} - z \right)^{\frac{3}{2}} - (w-z)^{\frac{3}{2}} \right] \psi(t) dt \right. \\ &\quad + \frac{2}{5} \int_{-1}^{nw} \left[\left(\frac{t}{n} - z \right)^{\frac{5}{2}} - (w-z)^{\frac{5}{2}} \right] \frac{\psi(t)}{z - \frac{t}{n}} dt - \int_{\frac{z}{n}}^w F_n(x, z) dx \Big\} \\ &\quad + \pi \int_{-1}^{nw} \psi(t) dt \left[(w-z)^{\frac{3}{2}} \int_{-1}^{nw} \frac{\hat{\psi}(t)}{z - \frac{t}{n}} dt + \hat{F}_n(w, z) \right] \\ &\quad - \pi \int_{-1}^{nw} \hat{\psi}(t) dt \left[(w-z)^{\frac{3}{2}} \int_{-1}^{nw} \frac{\psi(t)}{z - \frac{t}{n}} dt + F_n(w, z) \right], \\ A_3(w, z) &= \left[(w-z)^{\frac{3}{2}} \int_{-1}^{nw} \frac{\hat{\psi}(t)}{z - \frac{t}{n}} dt + \hat{F}_n(w, z) \right] \\ &\quad \times \left\{ \pi \int_{-1}^{nw} \left[\left(\frac{t}{n} - z \right)^{\frac{3}{2}} - (w-z)^{\frac{3}{2}} \right] \psi(t) dt \right. \\ &\quad + \frac{2}{5} \int_{-1}^{nw} \left[\left(\frac{t}{n} - z \right)^{\frac{5}{2}} - (w-z)^{\frac{5}{2}} \right] \frac{\psi(t)}{z - \frac{t}{n}} dt - \int_{\frac{z}{n}}^w F_n(x, z) dx \Big\} \\ &\quad - \left[(w-z)^{\frac{3}{2}} \int_{-1}^{nw} \frac{\psi(t)}{z - \frac{t}{n}} dt + F_n(w, z) \right] \\ &\quad \times \left\{ \pi \int_{-1}^{nw} \left[\left(\frac{t}{n} - z \right)^{\frac{3}{2}} - (w-z)^{\frac{3}{2}} \right] \hat{\psi}(t) dt \right. \\ &\quad + \frac{2}{5} \int_{-1}^{nw} \left[\left(\frac{t}{n} - z \right)^{\frac{5}{2}} - (w-z)^{\frac{5}{2}} \right] \frac{\hat{\psi}(t)}{z - \frac{t}{n}} dt - \int_{\frac{z}{n}}^w \hat{F}_n(x, z) dx \Big\}. \end{aligned} \right\} \quad (2.27)$$

and

$$\left. \begin{aligned} |A_2(w, z)| &\leq O\left(|\psi(nw)| + \left|\int_{-1}^{nw} \psi(t) dt\right|\right), \\ |A_3(w, z)| &\leq \frac{C}{n}. \end{aligned} \right\} \quad (2.28)$$

To do this, we recall that the estimates (2.20) and (2.21) on $\bar{S}_w$ in (O):

$$|F_n(w, z)| \leq O, \quad \left|\int_n^w F_n(x, z) dx\right| \leq \frac{C}{n},$$

and notice that

$$\begin{aligned} &\left| (w-z)^{\frac{3}{2}} \int_{-1}^{nw} \frac{\psi(t)}{z-\frac{t}{n}} dt \right| = \left| (w-z)^{\frac{1}{2}} \left(1-\frac{w}{z}\right) \int_{-1}^{nw} \frac{\psi(t)}{1-\frac{t}{nz}} dt \right| \leq O, \\ &\left| \int_{-1}^{nw} \left[\left(\frac{t}{n}-z\right)^{\frac{3}{2}} - (w-z)^{\frac{3}{2}} \right] \psi(t) dt \right| \\ &\quad \leq \frac{3}{2} \left| \int_{-1}^{nw} \left(\frac{t}{n}-w\right) \psi(t) dt \int_0^1 \left(\frac{t}{n}-z+\theta\left(w-\frac{t}{n}\right)\right)^{\frac{1}{2}} d\theta \right| \leq \frac{C}{n}, \\ &\left| \int_{-1}^{nw} \left[\left(\frac{t}{n}-z\right)^{\frac{5}{2}} - (w-z)^{\frac{5}{2}} \right] \frac{\psi(t)}{z-\frac{t}{n}} dt \right| \\ &\quad \leq \frac{5}{2} \left| \int_{-1+*}^{nw} \left(\frac{t}{n}-w\right) \psi(t) dt \int_0^1 \left(\frac{t}{n}-z+\theta\left(w-\frac{t}{n}\right)\right)^{\frac{1}{2}} \left(1+\theta \frac{nw-t}{t-nz}\right) d\theta \right| \\ &\quad \leq \frac{C}{n}, \\ &\int_{-1}^{nw} \hat{\psi}(t) dt = nw \int_{-1}^{nw} \psi(t) dt. \end{aligned}$$

We immediately get the first formula of (2.27) and the estimate (2.28).

On $\bar{S}_n$,

$$\begin{aligned} \eta_n(w, z) &= -\frac{3}{4} \{n(w-z)^{\frac{3}{2}} \tilde{\psi}(nz) + F_n(w, z)\}, \\ \bar{\tau}_n(w, z) &= \frac{15}{32} \left\{ (w-z)^{\frac{3}{2}} \int_{\infty}^{nz} \tilde{\psi}(y) dy + (w-z)^{\frac{3}{2}} \int_{nw}^{\infty} \tilde{\psi}(y) dy \right. \\ &\quad \left. + \int_{nz}^{nw} \left[(w-z)^{\frac{3}{2}} - \left(w-\frac{y}{n}\right)^{\frac{3}{2}} \right] \tilde{\psi}(y) dy - \int_n^w F_n(w, y) dy \right\}, \end{aligned}$$

where $\tilde{\psi}$ is the Hilbert transformation of ψ , that is,

$$\tilde{\psi}(z) = \int_{-1}^1 \frac{\psi(t)}{z-t} dt,$$

then

$$B_n = \eta_n \hat{\tau}_n - \hat{\eta}_n \bar{\tau}_n = -\frac{45}{128} \{n(w-z)^3 B_1(nz) + (w-z)^{\frac{3}{2}} B_2(w, z) + B_3(w, z)\}, \quad (2.29)$$

where

$$\begin{aligned}
 B_1(nz) &= \tilde{\psi}(nz) \int_{\infty}^{nz} \tilde{\psi}(y) dy - \tilde{\psi}(nz) \int_{\infty}^{nz} \tilde{\psi}(y) dy = \left(\int_{\infty}^{nz} \tilde{\psi}(y) dy \right)^2, \\
 B_2(w, z) &= n\tilde{\psi}(nz) \left\{ (w-z)^{\frac{3}{2}} \int_{nw}^{\infty} \tilde{\psi}(y) dy + \int_{nz}^{nw} \left[(w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n} \right)^{\frac{3}{2}} \right] \right. \\
 &\quad \times \tilde{\psi}(y) dy - \int_n^w \hat{F}_n(w, y) dy \Big\} - n\tilde{\psi}(nz) \left\{ (w-z)^{\frac{3}{2}} \int_{nw}^{\infty} \tilde{\psi}(y) dy \right. \\
 &\quad + \int_{nz}^{nw} \left[(w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n} \right)^{\frac{3}{2}} \right] \tilde{\psi}(y) dy - \int_n^w F_n(w, y) dy \Big\} \\
 &\quad + F_n(w, z) \int_{\infty}^{nz} \tilde{\psi}(y) dy - \hat{F}_n(w, z) \int_{\infty}^{nz} \tilde{\psi}(y) dy, \\
 B_3(w, z) &= F_n(w, z) \left\{ (w-z)^{\frac{3}{2}} \int_{nw}^{\infty} \tilde{\psi}(y) dy + \int_{nz}^{nw} \left[(w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n} \right)^{\frac{3}{2}} \right] \right. \\
 &\quad \times \tilde{\psi}(y) dy - \int_n^w \hat{F}_n(w, y) dy \Big\} - \hat{F}_n(w, z) \left\{ (w-z)^{\frac{3}{2}} \int_{nw}^{\infty} \tilde{\psi}(y) dy \right. \\
 &\quad + \int_{nz}^{nw} \left[(w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n} \right)^{\frac{3}{2}} \right] \tilde{\psi}(y) dy - \int_n^w F_n(w, y) dy \Big\},
 \end{aligned} \tag{2.30}$$

and

$$\begin{aligned}
 |B_2(w, z)| &\leq O n^{\frac{1}{2}} ((w-z) + n^{-1}), \\
 |B_3(w, z)| &\leq O n^{-\frac{1}{4}}.
 \end{aligned} \tag{2.31}$$

Above all, we show the first formula of (2.30). Denote

$$\varphi(t) = \int_{-1}^t \psi(s) ds,$$

$$\text{then } \tilde{\varphi}(y) = \int_{-1}^1 \frac{\varphi(t)}{y-t} dt \quad \text{and} \quad \tilde{\varphi}'(y) = \int_{-1}^1 \frac{\varphi'(t)}{y-t} dt = \tilde{\psi}(y).$$

Hence

$$\tilde{\varphi}(y) = \int_{\infty}^y \tilde{\psi}(y) dy.$$

Thus

$$\begin{aligned}
 \tilde{\psi}(y) &= \int_{-1}^1 \frac{t\psi(t)}{y-t} dt + \int_{-1}^1 \frac{\psi(s)ds}{y-t} dt = y \int_{-1}^1 \frac{\psi(t)}{y-t} dt + \int_{-1}^1 \frac{\psi(s)ds}{y-t} dt \\
 &= y\tilde{\psi}(y) + \tilde{\varphi}(y) = y\tilde{\psi}(y) + \int_{\infty}^y \tilde{\psi}(s) ds.
 \end{aligned}$$

It follows that

$$\begin{aligned}
 \int_{\infty}^{\infty} \tilde{\psi}(y) dy &= \int_{\infty}^{\infty} (y\tilde{\psi}(y) + \int_{\infty}^y \tilde{\psi}(s) ds) dy = \int_{\infty}^{\infty} y\tilde{\psi}(y) dy + \int_{\infty}^{\infty} \tilde{\psi}(s) ds \int_{\infty}^{\infty} dy \\
 &= \int_{\infty}^{\infty} y\tilde{\psi}(y) dy + \int_{\infty}^{\infty} (s-x)\tilde{\psi}(s) ds = x \int_{\infty}^{\infty} \tilde{\psi}(y) dy,
 \end{aligned}$$

and the first formula of (2.29) holds.

Moreover,

$$\begin{aligned}
 & \left| \int_s^w F_1^n(w, y) dy \right| \\
 &= \frac{2}{3} \left| \int_{-1}^1 \psi(t) \left[\int_0^{\frac{w-\chi(z, \frac{t}{n})}{w-\frac{t}{n}}} u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} \left(w-z-\left(w-\frac{t}{n}\right)u \right)^{\frac{3}{2}} du \right. \right. \\
 &\quad \left. \left. - \int_0^{\frac{w-\chi(z, \frac{t}{n})}{w-\frac{t}{n}}} u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} (w-z-wu)^{\frac{3}{2}} du \right] dt \right. \\
 &\quad \left. + 3 \int_{-1}^1 \psi(t) dt \int_0^{\frac{w-\chi(z, \frac{t}{n})}{w-\frac{t}{n}}} u^{-\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \right. \\
 &\quad \left. \times \int_0^1 \frac{\left(\left(w-\frac{t}{n} \right) (1-u) + \theta \left(\frac{t}{n} - z \right) \right)^{\frac{3}{2}} - \left(\left(w-\frac{t}{n} \right) (1-u) \right)^{\frac{3}{2}}}{\theta} d\theta \right. \\
 &\quad \left. + 3 \int_0^1 \frac{1-(1-\theta)^{\frac{3}{2}}}{\theta} d\theta \int_{-1}^1 \psi(t) \left[\int_0^{\frac{w-\chi(z, \frac{t}{n})}{w-\frac{t}{n}}} u^{-\frac{1}{2}} (1-u) \left(w-\frac{t}{n} \right)^{\frac{3}{2}} du \right. \right. \\
 &\quad \left. \left. - \int_0^{\frac{w-\chi(z, 0)}{w}} u^{-\frac{1}{2}} (1-u) w^{\frac{3}{2}} du \right] dt \right. \\
 &\quad \left. - 4 \int_{-1}^1 \left[\left(w-\chi \left(z, \frac{t}{n} \right) \right)^{\frac{3}{2}} - \left(w-\chi(z, 0) \right)^{\frac{3}{2}} \right] \psi(t) dt \right. \\
 &\quad \left. + \int_{ns}^1 \left(\frac{t}{n} - z \right) \psi(t) dt \int_0^1 d\theta \int_0^1 (w-z-(\theta+(1-\theta)\tau) \left(\frac{t}{n} - z \right))^{\frac{1}{2}} d\tau \right| \\
 &\leq \frac{2}{3} \left\{ \frac{1}{n} \int_{-1}^1 |t\psi(t)| dt \int_0^{\frac{w-\chi(z, \frac{t}{n})}{w-\frac{t}{n}}} u^{\frac{3}{2}} (1-u)^{-\frac{1}{2}} du \int_0^1 \left(w-z-wu+\frac{t}{n} u \right)^{\frac{1}{2}} d\tau \right. \\
 &\quad \left. + w^{\frac{3}{2}} \int_{-1}^1 |\psi(t)| dt \left| \int_{\frac{w-\chi(z, 0)}{w}}^{\frac{w-\chi(z, \frac{t}{n})}{w-\frac{t}{n}}} u^{\frac{1}{2}} (1-u) du \right| \right. \\
 &\quad \left. + \frac{3}{2} |z| \int_{-1}^1 |\psi(t)| dt \int_0^1 u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \int_0^1 (w(1-u)-z\theta)^{\frac{1}{2}} d\theta \right. \\
 &\quad \left. + \frac{9}{2} \int_{-1}^1 \left| \frac{t}{n} - z \right| |\psi(t)| dt \int_0^1 u^{-\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \int_0^1 d\theta \right. \\
 &\quad \left. \times \int_0^1 \left[\left(w-\frac{t}{n} \right) (1-u) + \theta \left(\frac{t}{n} - z \right) \tau \right]^{\frac{1}{2}} d\tau \right. \\
 &\quad \left. + \frac{C}{n} \int_{-1}^1 |t\psi(t)| dt \left[\frac{3}{2} \int_0^1 u^{-\frac{1}{2}} (1-u) du \int_0^1 \left(w-\frac{t}{n} \theta \right)^{\frac{1}{2}} d\theta \right. \right. \\
 &\quad \left. \left. + w^{\frac{3}{2}} \left| \left(\frac{w-\chi(z, 0)}{w} \right)^{\frac{1}{2}} - \left(\frac{w-\chi(z, \frac{t}{n})}{w-\frac{t}{n}} \right)^{\frac{1}{2}} \right| \right] dt \right\}
 \end{aligned}$$

$$\begin{aligned}
& -6 \int_{-1}^1 \left| \left(x(z, 0) - x\left(z, \frac{t}{n}\right) \right) \right| |\psi(t)| dt \\
& \times \int_0^1 \left(w - x(z, 0) + \theta \left(x(z, 0) - x\left(z, \frac{t}{n}\right) \right) \right)^{\frac{1}{2}} d\theta \\
& + \frac{1}{n} \int_{na}^1 |t - nz| |\psi(t)| dt \int_0^1 d\theta \int_0^1 \left(w - z - (\theta + (1-\theta)\tau) \left(\frac{t}{n} - z \right) \right)^{\frac{1}{2}} d\tau \Big\} \\
& \leq O \left\{ \frac{(w-z)^{\frac{1}{2}}}{n} + w^{\frac{1}{2}} \int_{-1}^1 |\psi(t)| \left(\left| \frac{x(z, 0) - x\left(z, \frac{t}{n}\right)}{1 - \frac{t}{nw}} \right| \right. \right. \\
& \quad \left. \left. + \frac{|t|}{n} \frac{1 - \frac{x(z, 0)}{w}}{1 - \frac{t}{nw}} \right) dt + \frac{w^{\frac{1}{2}}}{n} + \frac{w}{\sqrt{n}} \right\} \leq O(n^{-\frac{1}{2}}(w-z) + n^{-\frac{3}{2}}), \\
& \left| \int_s^w F_n^*(w, y) dy \right| \leq \left| \int_{-1}^1 \left(w - \frac{t}{n} \right) \psi(t) dt \int_0^1 u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \right. \\
& \quad \times \left| \frac{\int_0^1 \left(\left(w - \frac{t}{n} \right) (1-u) + \theta \left(\frac{t}{n} - z \right) \right)^{\frac{1}{2}} - \left(\left(w - \frac{t}{n} \right) (1-u) \right)^{\frac{1}{2}} \frac{\left| \frac{t}{n} - z \right|^{\frac{1}{2}}}{\theta^{\frac{1}{2}}} d\theta}{\left(\theta \left| \frac{t}{n} - z \right| \right)^{\frac{1}{2}}} \right. \\
& \quad \left. + 2 \int_0^1 \frac{1 - (1-\theta)^{\frac{1}{2}}}{\theta} d\theta \int_{-1}^1 |\psi(t)| \left| \int_0^{\frac{w-x(z, \frac{t}{n})}{w - \frac{t}{n}}} u^{\frac{1}{2}} \left(w - \frac{t}{n} \right)^{\frac{3}{2}} du \right. \right. \\
& \quad \left. \left. - \int_0^{\frac{w-x(z, 0)}{w}} u^{\frac{1}{2}} w^{\frac{3}{2}} du \right| dt \right| \\
& \leq O \left\{ \frac{w + \frac{1}{n}}{n^{\frac{1}{2}}} + \frac{1}{n} \int_{-1}^1 |t \psi(t)| dt \int_0^1 u^{\frac{1}{2}} du \int_0^1 \left(w - \theta \frac{t}{n} \right)^{\frac{1}{2}} d\theta \right. \\
& \quad \left. + w^{\frac{1}{2}} \int_{-1}^1 |\psi(t)| \left(\left| x(z, 0) \right| + \frac{\left| x\left(z, \frac{t}{n}\right) - \frac{t}{n} \right|}{1 - \frac{t}{nw}} \right) dt \right\} \\
& \leq O \left(\frac{w + \frac{1}{n}}{\sqrt{n}} + \frac{\sqrt{w}}{n} \right) \leq O(n^{-\frac{1}{2}}(w-z) + n^{-\frac{3}{2}}).
\end{aligned}$$

So

$$\left| \int_s^w F_n(w, y) dy \right| \leq O(n^{-\frac{1}{2}}(w-z) + n^{-\frac{3}{2}}).$$

Recall that the estimate of $F_n(w, z)$ on the $\bar{S}_s$ in (O):

$$|F_n(w, z)| \leq O \min(n^{\frac{1}{2}}(w-z) + n^{-\frac{1}{2}}, n^{\frac{1}{4}}(w-z)^{\frac{3}{4}} + n^{-\frac{1}{2}}),$$

and remark that

$$\begin{aligned}
 \left| (w-z)^{\frac{3}{2}} \int_{nw}^{\infty} \tilde{\psi}(y) dy \right| &= (w-z)^{\frac{3}{2}} \left| \int_{-1}^1 t \psi(t) dt \int_{nw}^{\infty} \frac{1}{y(y-t)} dy \right| \\
 &\leq (w-z)^{\frac{3}{2}} \int_{-1+\varepsilon}^{1-\varepsilon} |t \psi(t)| dt \int_{nw}^{\infty} \frac{dy}{(y-1+\varepsilon)^2} \\
 &\leq C n^{-\frac{1}{2}} (w-z) \left(1 - \frac{z}{w}\right)^{\frac{1}{2}} \left(1 - \frac{1-\varepsilon}{nw}\right)^{-\frac{1}{2}} \\
 &\leq C n^{-\frac{1}{2}} (w-z), \\
 \left| \int_{nw}^{nw} \left[(w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n}\right)^{\frac{3}{2}} \right] \tilde{\psi}(y) dy \right| \\
 &= \frac{3}{2} \left| \int_{nw}^1 n^{-\frac{1}{2}} (y-nz)^{\frac{1}{2}} (w-z)^{\frac{1}{2}} |\tilde{\psi}(y)| dy \int_0^1 \left(w - \frac{y}{n} + \theta \left(\frac{y}{n} - z\right)\right)^{\frac{1}{2}} d\theta \right| \\
 &\quad + \left| \int_1^{nw} \tilde{\psi}(y) (w-z) \frac{1 - \left(\frac{w - \frac{y}{n}}{w-z}\right)^{\frac{3}{2}}}{\left(\frac{\frac{y}{n} - z}{w-z}\right)^{\frac{1}{2}}} \left(\frac{y}{n} - z\right)^{\frac{1}{2}} dy \right| \\
 &\leq C n^{-\frac{1}{2}} (w-z) + n^{-\frac{1}{2}} (w-z) \int_{-1}^1 |t \psi(t)| dt \int_1^{nw} \frac{\left(1 - \frac{nz}{y}\right)^{\frac{1}{2}}}{\sqrt{y(y-t)}} dy \\
 &\leq C n^{-\frac{1}{2}} (w-z), \\
 \left| \int_{\infty}^{nw} \tilde{\psi}(y) dy \right| &\leq \int_{nw}^1 \left| \int_{-1}^1 \frac{\psi(y) - \psi(t)}{y-t} dt + \psi(y) \ln \frac{1-y}{1+y} \right| dy \\
 &\quad + \left| \int_1^{\infty} dy \int_{-1}^1 \frac{t \psi(t)}{y(y-t)} dt \right| \leq O.
 \end{aligned}$$

It follows that (2.30) holds.

On I , recall that $|\eta_n| + |\tau_n| \leq C n^{-\frac{1}{2}}$, then $|B_n| \leq C n^{-1}$.

On U_1 , observe that $\eta_n(w, z) \equiv 0$, then $B_n \equiv 0$.

On Q ,

$$\begin{aligned}
 \eta_n(w, z) &= -\frac{3}{4} \{n(w-z)^{\frac{3}{2}} \tilde{\psi}(nz) + F_n(w, z)\}, \\
 \tau_n(w, z) &= -\frac{15}{32} \left\{ -\frac{2}{5} n(w-z)^{\frac{5}{2}} \tilde{\psi}(nz) + \left(\pi - \frac{2}{5}\right) \int_{-1}^1 \left(\frac{t}{n} - z\right)^{\frac{3}{2}} \psi(t) dt \right. \\
 &\quad \left. - \int_z^w F_n(x, z) dx \right\}, \\
 \bar{\tau}_n(w, z) &= \frac{15}{32} \left\{ (w-z)^{\frac{3}{2}} \int_{\infty}^{nw} \tilde{\psi}(y) dy + (w-z)^{\frac{3}{2}} \int_{nw}^{\infty} \tilde{\psi}(y) dy \right. \\
 &\quad \left. + \int_{nw}^{nw} \left[(w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n}\right)^{\frac{3}{2}} \right] \tilde{\psi}(y) dy - \int_z^w F_n(w, y) dy \right\}.
 \end{aligned}$$

Divide Q into two parts $Q = Q_1 \cup Q_2$,

$$Q_1 = \{(w, z): w > -Lz\} \cap Q, \quad Q_2 = \{(w, z): w \leq -Lz\} \cap Q,$$

where L is a sufficiently large constant.

On the triangular zone Q_1 ,

$$B_n = -\frac{45}{128} \{n(w-z)^{\frac{3}{2}} [C_1(nz)(w-z)^{\frac{3}{2}} + C_2(w, z)] \\ + (w-z)^{\frac{3}{2}} C_3(w, z) + C_4(w, z)\}, \quad (2.32)$$

where

$$\left. \begin{aligned} C_1(nz) &= \tilde{\psi}(nz) \int_{-\infty}^{nz} \tilde{\psi}(y) dy - \tilde{\psi}(nz) \int_{-\infty}^{nz} \hat{\psi}(y) dy = \left(\int_{-\infty}^{nz} \tilde{\psi}(y) dy \right)^2, \\ C_2(w, z) &= \tilde{\psi}(nz) \left\{ \int_{nz}^{nw} \left[(w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n} \right)^{\frac{3}{2}} \right] \tilde{\psi}(y) dy - \int_s^w \hat{F}_n(w, y) dy \right\} \\ &\quad - \tilde{\psi}(nz) \left\{ \int_{nz}^{nw} \left[(w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n} \right)^{\frac{3}{2}} \right] \tilde{\psi}(y) dy - \int_s^w F_n(w, y) dy \right\}, \\ C_3(w, z) &= n(w-z)^{\frac{3}{2}} \tilde{\psi}(nz) \int_{nw}^{\infty} \tilde{\psi}(y) dy + F_n(w, z) \int_{-\infty}^{nz} \tilde{\psi}(y) dy \\ &\quad - n(w-z)^{\frac{3}{2}} \tilde{\psi}(nz) \int_{nw}^{\infty} \tilde{\psi}(y) dy - \hat{F}_n(w, z) \int_{-\infty}^{nz} \tilde{\psi}(y) dy, \\ C_4(w, z) &= F_n(w, z) \left\{ (w-z)^{\frac{3}{2}} \int_{nw}^{\infty} \tilde{\psi}(y) dy \right. \\ &\quad + \int_{nz}^{nw} \left[(w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n} \right)^{\frac{3}{2}} \right] \tilde{\psi}(y) dy - \int_s^w \hat{F}_n(w, y) dy \Big\} \\ &\quad - \hat{F}_n(w, z) \left[(w-z)^{\frac{3}{2}} \int_{nw}^{\infty} \tilde{\psi}(y) dy \right. \\ &\quad + \left. \left. \int_{nz}^{nw} \left[(w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n} \right)^{\frac{3}{2}} \right] \tilde{\psi}(y) dy - \int_s^w F_n(w, y) dy \right\} \right] \end{aligned} \right\} \quad (2.33)$$

and

$$\left. \begin{aligned} (w-z)^{\frac{3}{2}} + \frac{C_2(w, z)}{C_1(nz)} &\geq C_0(w-z)^{\frac{3}{2}} - \frac{C(w-z)}{\sqrt{n}}, \quad C_0 > 0 \text{ constant}, \\ |C_3(w, z)| &\leq C(n^{\frac{1}{2}}(w-z)C_1(nz) + n^{-\frac{1}{2}}), \\ |C_4(w, z)| &\leq Cn^{-\frac{1}{4}}. \end{aligned} \right\} \quad (2.34)$$

The first formula of (2.33) may be proved similarly to that of (2.30). Furthermore,

$$\begin{aligned}
& \left| \int_z^u F_1^n(w, y) dy \right| \\
& \leq \frac{2}{3} \left\{ \left| \int_{-1}^1 \psi(t) dt \int_0^1 u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} \left[\left(w-z - \left(w-\frac{t}{n} \right) u \right)^{\frac{3}{2}} - (w-z-wu)^{\frac{3}{2}} \right] du \right| \right. \\
& \quad + \frac{9}{2} \left| \int_{-1}^1 \psi(t) dt \int_0^1 u^{-\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \int_0^1 d\theta \int_0^1 \left\{ \left[(1-\theta)(1-u) \left(w-\frac{t}{n} \right) \right. \right. \right. \\
& \quad \left. \left. \left. + \theta \tau \left(w-z - \left(w-\frac{t}{n} \right) u \right) \right]^{\frac{1}{2}} \left(w-z - \left(w-\frac{t}{n} \right) u \right) \right. \right. \\
& \quad \left. \left. - \left[(1-\theta)(1-u)w + \theta \tau (w-z-wu) \right]^{\frac{1}{2}} (w-z-wu) \right\} d\tau \right| \\
& \quad + \frac{6}{n} \int_{-1}^1 |t\psi(t)| dt \int_0^1 \left(w-\frac{t}{n} \theta \right)^{\frac{1}{2}} d\theta + \int_{-1}^1 |\psi(t)| dt \int_0^1 d\theta \\
& \quad \times \int_0^1 \left| \left(\frac{t}{n} - z \right) \left(w-z - (\theta + (1-\theta)\tau) \left(\frac{t}{n} - z \right) \right)^{\frac{1}{2}} \right. \\
& \quad \left. + z \left(w-z + (\theta + (1-\theta)\tau)z \right)^{\frac{1}{2}} \right| d\tau \Big\} \\
& = \frac{2}{3} \left\{ \frac{3}{2n} \int_{-1}^1 |t\psi(t)| dt \int_0^1 u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \int_0^1 \left(w-z - \left(w-\frac{t}{n} \theta \right) u \right)^{\frac{1}{2}} \\
& \quad + \frac{9}{2n} \int_{-1}^1 |t\psi(t)| dt \int_0^1 u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \int_0^1 d\theta \\
& \quad \times \int_0^1 (w-z)^{\frac{1}{2}} \left[(1-\theta)(1-u) \frac{w}{w-z} + \theta \tau \left(1 - \frac{w}{w-z} u \right) \right]^{\frac{1}{2}} d\tau \\
& \quad + \frac{9}{2} \int_{-1}^1 |\psi(t)| dt \int_0^1 u^{-\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \int_0^1 d\theta \int_0^1 \left(w-z - \left(w-\frac{t}{n} \right) u \right) \\
& \quad \times \left| \frac{\left\{ \left[(1-\theta)(1-u) \left(w-\frac{t}{n} \right) + \theta \tau \left(w-z - \left(w-\frac{t}{n} \right) u \right) \right]^{\frac{1}{2}} \right.}{\left. - \left[(1-\theta)(1-u)w + \theta \tau (w-z-wu) \right]^{\frac{1}{2}} \right\}}{\left[(\theta \tau - (1-\theta)(1-u)) \frac{|t|}{n} \right]^{\frac{1}{2}}} \right| \\
& \quad \times \left[(\theta \tau - (1-\theta)(1-u)) \frac{|t|}{n} \right]^{\frac{1}{2}} d\tau + \frac{6}{n} (w-z)^{\frac{1}{2}} \int_{-1}^1 |t\psi(t)| dt \\
& \quad \times \int_0^1 \left(\frac{w-\frac{t}{n} \theta}{w-z} \right)^{\frac{1}{2}} d\theta + \frac{1}{n} \int_{-1}^1 |t\psi(t)| dt \int_0^1 d\theta \\
& \quad \times \int_0^1 \left(w-z - (\theta + (1-\theta)\tau) \left(\frac{t}{n} - z \right) \right)^{\frac{1}{2}} d\tau + (-z) \int_{-1}^1 |\psi(t)| dt \int_0^1 d\theta \\
& \quad \times \int_0^1 \left| \frac{\left(w-z - (\theta + (1-\theta)\tau) \left(\frac{t}{n} - z \right) \right)^{\frac{1}{2}} - (w-z + (\theta + (1-\theta)\tau)z)^{\frac{1}{2}}}{\left[(\theta + (1-\theta)\tau) \frac{|t|}{n} \right]^{\frac{1}{2}}} \right| \\
& \quad \left. \right\}
\end{aligned}$$

$$\begin{aligned}
& \times \left[(\theta + (1-\theta)\tau) \frac{|t|}{n} \right]^{\frac{1}{2}} d\tau \} \leq O \left\{ \frac{(w-z)^{\frac{1}{2}}}{n} + \frac{w-z}{\sqrt{n}} + \frac{-z}{\sqrt{n}} \right\} \\
& \leq O n^{-\frac{1}{2}} (w-z). \\
& \left| \int_w^w F_n^2(w, y) dy \right| = 2 \left| \int_{-1}^1 \psi(t) dt \int_0^1 u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \right. \\
& \quad \times \int_0^1 \left(w - \frac{t}{n} \right) \left[\frac{\left(\left(w - \frac{t}{n} \right) (1-u) + \theta \left(\frac{t}{n} - z \right) \right)^{\frac{1}{2}} - \left(\left(w - \frac{t}{n} \right) (1-u) \right)^{\frac{1}{2}}}{\theta} \right. \\
& \quad \left. \left. + (1-u)^{\frac{1}{2}} \frac{1-(1-\theta)^{\frac{1}{2}}}{1-\theta} \left(w - \frac{t}{n} \right)^{\frac{1}{2}} \right] d\theta \right| \\
& \leq 2 \int_{-1}^1 |\psi(t)| dt \int_0^1 u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} du \\
& \quad \times \int_0^1 \left(w - \frac{t}{n} \right) \left| \frac{\left[\left(w - \frac{t}{n} \right) (1-u) + \theta \left(\frac{t}{n} - z \right) \right]^{\frac{1}{2}} - \left[\left(w - \frac{t}{n} \right) (1-u) \right]^{\frac{1}{2}}}{\left(\frac{t}{n} - z \right)^{\frac{1}{2}}} \right| \\
& \quad \times \frac{\left(\frac{t}{n} - z \right)^{\frac{1}{2}}}{\theta^{\frac{1}{2}}} d\theta + 2 \int_0^1 \frac{1-(1-\theta)^{\frac{1}{2}}}{1-\theta} d\theta \\
& \quad \times \int_0^1 u^{\frac{1}{2}} du \left| \int_{-1}^1 \psi(t) \left(\left(w - \frac{t}{n} \right)^{\frac{3}{2}} - w^{\frac{3}{2}} \right) dt \right| \\
& \leq O \left[\frac{w-z}{\sqrt{n}} + (w-z)^{\frac{3}{2}} \left(-\frac{z}{w} \right)^{\frac{1}{2}} + \frac{w-z}{\sqrt{n} (nw-nz)} \right] \\
& \leq O \left(n^{-\frac{1}{2}} (w-z) + (w-z)^{\frac{3}{2}} \left(-\frac{z}{w} \right)^{\frac{1}{2}} \right).
\end{aligned}$$

Hence

$$\left| \int_w^w F_n(w, y) dy \right| \leq O \left(n^{-\frac{1}{2}} (w-z) + (w-z)^{\frac{3}{2}} \left(-\frac{z}{w} \right)^{\frac{1}{2}} \right). \quad (2.35)$$

Also,

$$\begin{aligned}
& \left| \int_{ns}^{nw} \left[(w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n} \right)^{\frac{3}{2}} \right] \tilde{\psi}(y) dy \right| \leq \int_{-1}^1 \left| (w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n} \right)^{\frac{3}{2}} \right| |\tilde{\psi}(y)| dy \\
& \quad + \left| \int_1^{nw} \left[(w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n} \right)^{\frac{3}{2}} \right] \int_{-1}^1 \frac{\psi(t)}{y-t} dt dy \right| \\
& \quad + \left| \int_{ns}^{-1} \left[(w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n} \right)^{\frac{3}{2}} \right] \int_{-1}^1 \frac{\psi(t)}{y-t} dt dy \right| \\
& \leq \frac{3}{2} \int_{-1}^1 \left(\frac{y}{n} - z \right) \int_0^1 \left(w - z + \theta \left(z - \frac{y}{n} \right) \right)^{\frac{1}{2}} d\theta |\tilde{\psi}(y)| dy
\end{aligned}$$

$$\begin{aligned}
& + \left| \int_1^{nw} \left[(w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n} \right)^{\frac{3}{2}} \right] \int_{-1}^1 \frac{\psi(t)}{y-t} dt dy \right. \\
& \quad \left. - \int_1^{-nz} \left[(w-z)^{\frac{3}{2}} - \left(w + \frac{y}{n} \right)^{\frac{3}{2}} \right] \int_{-1}^1 \frac{\psi(t)}{y+t} dt dy \right| \\
& \leq (w-z)^{\frac{1}{2}} \frac{1-nz}{n} O + \left| \int_{-1}^1 t\psi(t) dt \int_1^{-nz} \left(\frac{(w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n} \right)^{\frac{3}{2}}}{y(y-t)} \right. \right. \\
& \quad \left. \left. - \frac{(w-z)^{\frac{3}{2}} - \left(w + \frac{y}{n} \right)^{\frac{3}{2}}}{y(y+t)} \right) dy \right| + \left| \int_{-nz}^{nw} \left[(w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n} \right)^{\frac{3}{2}} \right] \int_{-1}^1 \frac{t\psi(t)}{y(y-t)} dt dy \right| \\
& \leq (w-z)^{\frac{1}{2}} \frac{1-nz}{n} O + \int_{-1}^1 |t\psi(t)| dt \int_1^{-nz} \left| \left(w + \frac{y}{n} \right)^{\frac{3}{2}} - \left(w - \frac{y}{n} \right)^{\frac{3}{2}} \right| \frac{dy}{y(y-t)} \\
& \quad + \int_{-1}^1 |t\psi(t)| dt \int_1^{-nz} \left| (w-z)^{\frac{3}{2}} - \left(w + \frac{y}{n} \right)^{\frac{3}{2}} \right| \frac{dy}{y(y^2-t^2)} \\
& \quad + \frac{3}{2} \int_{-1}^1 |t\psi(t)| dt \int_{-nz}^{nw} \left(-z + \frac{y}{n} \right) \int_0^1 \left(w - z + \theta \left(z - \frac{y}{n} \right) \right)^{\frac{1}{2}} d\theta \frac{dy}{y(y-t)} \\
& \leq O \left\{ (w-z)^{\frac{1}{2}} \frac{1-nz}{n} + (-z)(w-z)^{\frac{1}{2}} \int_{-1}^1 |t\psi(t)| dt \int_1^{-nz} \frac{dy}{y(y-t)} \right. \\
& \quad + (-z)(w-z)^{\frac{1}{2}} \int_{-1}^1 |t\psi(t)| dt \int_1^{-nz} \frac{dy}{y(y^2-t^2)} \\
& \quad \left. + n^{-\frac{1}{2}}(w-z) \int_{-1}^1 |t\psi(t)| dt \int_{-nz}^{nw} \frac{dy}{\sqrt{y(y-t)}} \right\} \\
& = O \left\{ (w-z)^{\frac{1}{2}} \frac{1-nz}{n} + (-z)(w-z)^{\frac{1}{2}} \int_{-1+s}^{1-s} |t\psi(t)| dt \int_1^{-nz} \frac{dy}{(y-1+s)^{\frac{3}{2}}} \right. \\
& \quad \left. + n^{-\frac{1}{2}}(w-z) \int_{-1}^1 |t\psi(t)| dt \int_{-nz}^{nw} \frac{dy}{(y-1+s)^{\frac{3}{2}}} \right\} \\
& \leq O \left\{ n^{-\frac{1}{2}}(w-z) + (w-z)^{\frac{3}{2}} \left(-\frac{z}{w} \right) \right\}.
\end{aligned}$$

Moreover, notice that

$$\left(\int_{-\infty}^x \tilde{\psi}(y) dy \right)^2 \geq O_M > 0, \quad x \in [1, M],$$

$$\lim_{x \rightarrow \infty} \frac{\tilde{\psi}(x)}{\left(\int_{-\infty}^x \tilde{\psi}(y) dy \right)^2} = \frac{1}{\int_{-1}^1 t\psi(t) dt},$$

$$\lim_{x \rightarrow \infty} \frac{\tilde{\psi}(x)}{\left(\int_{-\infty}^x \tilde{\psi}(y) dy \right)^2} = \frac{\int_{-1}^1 t\hat{\psi}(t) dt}{\left(\int_{-1}^1 t\psi(t) dt \right)^2}.$$

So

$$\left| \frac{\tilde{\psi}(nz)}{\left(\int_{-\infty}^{nz} \tilde{\psi}(y) dy \right)^2} \right| + \left| \frac{\tilde{\psi}(nz)}{\left(\int_{nz}^{\infty} \tilde{\psi}(y) dy \right)^2} \right| \leq O, \quad -nz > 1.$$

It follows from the above estimates (2.17) and (2.35) that

$$\left| \frac{O_2(w, z)}{(w-z)O_1(nz)} \right| \leq O\left(\frac{1}{\sqrt{n}} + (w-z)^{\frac{1}{2}}\left(-\frac{z}{w}\right)^{\frac{1}{2}}\right).$$

Thus

$$\begin{aligned} (w-z)^{\frac{3}{2}} + \frac{O_2(w, z)}{O_1(nz)} &\geq (w-z) \left[(w-z)^{\frac{1}{2}} - O\left(\frac{1}{\sqrt{n}} + (w-z)^{\frac{1}{2}}\left(-\frac{z}{w}\right)^{\frac{1}{2}}\right) \right] \\ &= (w-z) \left[(w-z)^{\frac{1}{2}} \left(1 - O\left(-\frac{z}{w}\right)^{\frac{1}{2}}\right) - \frac{O}{\sqrt{n}} \right] \\ &\geq O_0(w-z)^{\frac{3}{2}} - \frac{O(w-z)}{\sqrt{n}}, \quad (L \text{ sufficiently large}). \end{aligned}$$

Also,

$$\begin{aligned} \left| (w-z)^{\frac{3}{2}} \int_{nw}^{\infty} \tilde{\psi}(y) dy \right| &\leq \frac{O}{n} (w-z)^{\frac{1}{2}} \left(1 - \frac{z}{w}\right) \left(1 - \frac{1-\varepsilon}{nw}\right)^{-1} \leq \frac{O(w-z)^{\frac{1}{2}}}{n}, \\ \left| \frac{1}{nz \int_{-\infty}^{nz} \tilde{\psi}(y) dy} \right| &\leq O, \quad -nz > 1, \end{aligned}$$

and the estimate (2.17) holds, i.e.

$$|F_n(w, z)| \leq O(n^{-\frac{1}{2}} + n^{\frac{1}{4}}(-nz)^{-\frac{3}{4}}w) \leq O(n^{-\frac{1}{2}} + n^{\frac{1}{2}}(-nz)^{-1}(w-z)).$$

We obtain

$$\begin{aligned} |O_3(w, z)| &\leq O\left\{[(w-z)^{\frac{1}{2}} + n^{\frac{1}{2}}(w-z)]O_1(nz) + n^{-\frac{1}{2}}\right\} \\ &\leq O(n^{\frac{1}{2}}(w-z)O_1(nz) + n^{-\frac{1}{2}}), \\ |O_4(w, z)| &\leq On^{-\frac{1}{4}}. \end{aligned}$$

This completes the proof of (2.34).

On the other triangular zone Q_2 , notice that

$$\begin{aligned} |n(w-z)^{\frac{3}{2}}\tilde{\psi}(nz)| &= \left| (w-z)^{\frac{3}{2}} \int_{-1}^1 \frac{\psi(t)}{z - \frac{t}{n}} dt \right| \\ &= \frac{(w-z)^{\frac{3}{2}}}{n} \left| \int_{-1}^1 \frac{t\psi(t)}{z\left(z - \frac{t}{n}\right)} dt \right| \\ &= \left| \frac{\left(1 - \frac{w}{z}\right)^{\frac{3}{2}}}{\sqrt{n}} \int_{-1+\varepsilon}^{1-\varepsilon} \frac{|t\psi(t)|}{\left(1 - \frac{t}{nz}\right)\sqrt{nz}} dt \right| \leq \frac{O}{\sqrt{n}}, \end{aligned}$$

$$\left| \int_{-1}^1 \left(\frac{t}{n} - z \right)^{\frac{3}{2}} \psi(t) dt \right| = \left| \int_{-1}^1 \left[\left(\frac{t}{n} - z \right)^{\frac{3}{2}} - (-z)^{\frac{3}{2}} \right] \psi(t) dt \right| \leq \frac{C}{n},$$

and the estimate (2.17) holds, i.e.

$$|F_n(w, z)| \leq C \left(n^{-\frac{1}{2}} + n^{-\frac{1}{4}} w^{\frac{1}{2}} \left(-\frac{w}{z} \right)^{\frac{1}{2}} \right) \leq C n^{-\frac{1}{4}},$$

$$\left| \int_0^w F_n(x, z) dx \right| \leq C \left(n^{-\frac{1}{2}} + n^{-\frac{1}{4}} w^{\frac{1}{2}} \left(-\frac{w}{z} \right)^{\frac{1}{2}} \right) \leq C n^{-\frac{1}{4}}.$$

Therefore,

$$|B_n| \leq C n^{-\frac{1}{2}}.$$

On U_2 ,

$$\eta_n(w, z) = -\frac{3}{4} \{ n(w-z)^{\frac{3}{2}} \tilde{\psi}(nz) + F_n(w, z) \},$$

$$\begin{aligned} \bar{\tau}_n(w, z) = & \frac{15}{32} \left\{ (w-z)^{\frac{3}{2}} \int_{\infty}^{nz} \tilde{\psi}(y) dy + (w-z)^{\frac{3}{2}} \int_{nw}^{\infty} \tilde{\psi}(y) dy \right. \\ & \left. + \int_{nz}^{nw} \left[(w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n} \right)^{\frac{3}{2}} \right] \tilde{\psi}(y) dy - \int_0^w F_n(w, y) dy \right\}, \end{aligned}$$

then

$$B_n = -\frac{45}{128} \{ n(w-z)^3 D_1(nz) + (w-z)^{\frac{3}{2}} D_2(w, z) + D_3(w, z) \}, \quad (2.36)$$

where

$$\left. \begin{aligned} D_1(nz) &= \tilde{\psi}(nz) \int_{\infty}^{nz} \tilde{\psi}(y) dy - \tilde{\psi}(nz) \int_{\infty}^{nz} \tilde{\psi}(y) dy = \left(\int_{\infty}^{nz} \tilde{\psi}(y) dy \right)^2, \\ D_2(w, z) &= n \tilde{\psi}(nz) \left[(w-z)^{\frac{3}{2}} \int_{nw}^{\infty} \tilde{\psi}(y) dy + \int_{nz}^{nw} \left[(w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n} \right)^{\frac{3}{2}} \right] \right. \\ &\quad \times \tilde{\psi}(y) dy + \int_0^w \hat{F}_n(w, y) dy \left. \right] + F_n(w, z) \int_{\infty}^{nz} \tilde{\psi}(y) dy \\ &\quad - n \tilde{\psi}(nz) \left[(w-z)^{\frac{3}{2}} \int_{nw}^{\infty} \tilde{\psi}(y) dy + \int_{nz}^{nw} \left[(w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n} \right)^{\frac{3}{2}} \right] \right. \\ &\quad \times \tilde{\psi}(y) dy + \int_0^w F_n(w, y) dy \left. \right] - \hat{F}_n(w, z) \int_{\infty}^{nz} \tilde{\psi}(y) dy, \\ D_3(w, z) &= F_n(w, z) \left[(w-z)^{\frac{3}{2}} \int_{nw}^{\infty} \tilde{\psi}(y) dy + \int_{nz}^{nw} \left[(w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n} \right)^{\frac{3}{2}} \right] \right. \\ &\quad \times \tilde{\psi}(y) dy - \int_0^w \hat{F}_n(w, y) dy \left. \right] - \hat{F}_n(w, z) \left[(w-z)^{\frac{3}{2}} \int_{nw}^{\infty} \tilde{\psi}(y) dy \right. \\ &\quad \left. + \int_{nz}^{nw} \left[(w-z)^{\frac{3}{2}} - \left(w - \frac{y}{n} \right)^{\frac{3}{2}} \right] \tilde{\psi}(y) dy - \int_0^w F_n(w, y) dy \right], \end{aligned} \right\} \quad (2.37)$$

and

$$\left. \begin{aligned} |D_2(w, z)| &\leq O(n^{\frac{1}{2}}(w-z)D_1(nz) + n^{-\frac{1}{2}}), \\ |D_3(w, z)| &\leq On^{-\frac{1}{4}}. \end{aligned} \right\} \quad (2.38)$$

In fact,

$$\begin{aligned} |F_1^n(w, z)| &\leq \left| \int_{-1}^1 \psi(t) dt \left[\int_0^{\frac{w-z}{w-\frac{t}{n}}} u^{\frac{1}{2}}(1-u)^{-\frac{1}{2}} \left(w-z - \left(w-\frac{t}{n} \right) u \right)^{\frac{1}{2}} du \right. \right. \\ &\quad \left. \left. - \int_0^{\frac{w-z}{w}} u^{\frac{1}{2}}(1-u)^{-\frac{1}{2}} (w-z-wu)^{\frac{1}{2}} du \right] + 3 \left| \int_{-1}^1 \psi(t) dt \left[\int_0^{\frac{w-z}{w-\frac{t}{n}}} u^{-\frac{1}{2}}(1-u)^{-\frac{1}{2}} \right. \right. \right. \\ &\quad \times \int_0^1 \left(\left(w-\frac{t}{n} \right) (1-u) + \theta \left(\frac{t}{n} - z \right) \right)^{\frac{1}{2}} d\theta du \\ &\quad \left. \left. - \int_0^{\frac{w-z}{w}} u^{-\frac{1}{2}}(1-u)^{-\frac{1}{2}} du \int_0^1 (w(1-u) - \theta z)^{\frac{1}{2}} d\theta \right] \right| \\ &\leq \int_{-1}^1 |\psi(t)| \left[\int_0^{\frac{w-z}{w-\frac{t}{n}}} u^{\frac{1}{2}}(1-u)^{-\frac{1}{2}} \left| \frac{\left(w-z - \left(w-\frac{t}{n} \right) u \right)^{\frac{1}{2}} - (w-z-wu)^{\frac{1}{2}}}{\left(\frac{|t|}{n} u \right)^{\frac{1}{2}}} \right| \right. \\ &\quad \times \left(\frac{|t|}{n} u \right)^{\frac{1}{2}} du + w^{\frac{1}{2}} \left| \int_0^{\frac{w-z}{w-\frac{t}{n}}} u^{\frac{1}{2}} du \right| + 3 \int_{-1}^1 |\psi(t)| \left[\int_0^{\frac{w-z}{w-\frac{t}{n}}} u^{-\frac{1}{2}}(1-u)^{-\frac{1}{2}} du \right. \\ &\quad \times \int_0^1 \left| \frac{\left(\left(w-\frac{t}{n} \right) (1-u) + \theta \left(\frac{t}{n} - z \right) \right)^{\frac{1}{2}} - (w(1-u) - \theta z)^{\frac{1}{2}}}{\left(|\theta+u-1| \frac{|t|}{n} \right)^{\frac{1}{2}}} \right| \\ &\quad \times \left(|\theta+u-1| \frac{|t|}{n} \right)^{\frac{1}{2}} d\theta + w^{\frac{1}{2}} \left| \int_0^{\frac{w-z}{w-\frac{t}{n}}} u^{-\frac{1}{2}} du \right| \left. \right] dt \\ &\leq O \left(\frac{1}{\sqrt{n}} + \frac{1}{n} \int_{-1+\varepsilon}^{1-\varepsilon} |t| \frac{w-z}{w-\frac{t}{n}} dt \right. \\ &\quad \left. + \int_{-1+\varepsilon}^{1-\varepsilon} \sqrt{\frac{w-z}{w-\frac{t}{n}}} \left| \frac{w^{\frac{1}{2}} - \left(w-\frac{t}{n} \right)^{\frac{1}{2}}}{\left(\frac{|t|}{n} \right)^{\frac{1}{2}}} \right| \left(\frac{|t|}{n} \right)^{\frac{1}{2}} \right) \leq \frac{C}{\sqrt{n}}, \\ |F_2^n(w, z)| &= \left| \int_{-1}^1 \psi(t) dt \int_0^{\frac{w-z}{w-\frac{t}{n}}} u^{\frac{1}{2}}(1-u)^{-\frac{1}{2}} du \right. \\ &\quad \times \int_0^1 \left(w-\frac{t}{n} \right) \left(\left(w-\frac{t}{n} \right) (1-u) + \theta \left(\frac{t}{n} - z \right) \right)^{-\frac{1}{2}} d\theta \left. \right| \end{aligned}$$

$$\begin{aligned}
&= \left| \int_{-1}^1 \psi(t) \left\{ \int_0^{\frac{w-z}{w-\frac{t}{n}}} u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} \left[\left(\left(w - \frac{t}{n} \right) (1-u) + \frac{t}{n} - z \right)^{\frac{1}{2}} \right. \right. \right. \\
&\quad \left. \left. - \left(\left(w - \frac{t}{n} \right) (1-u) \right)^{\frac{1}{2}} \right] \frac{w - \frac{t}{n}}{z - \frac{t}{n}} du \right. \right. \\
&\quad \left. \left. - \int_0^{\frac{w-z}{w}} u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} \left[(w(1-u) - z)^{\frac{1}{2}} - (w(1-u))^{\frac{1}{2}} \right] \frac{w}{z} du \right\} dt \right| \\
&= \int_{-1}^1 |\psi(t)| dt \left| \int_{\frac{w-z}{n}}^{\frac{w-z}{w-\frac{t}{n}}} u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} \left| \frac{(w(1-u) - z)^{\frac{1}{2}} - (w(1-u))^{\frac{1}{2}}}{z^{\frac{1}{2}}} \right| \frac{w}{z^{\frac{1}{2}}} du \right| \\
&\quad + \int_{-1}^1 |\psi(t)| dt \left| \int_0^{\frac{w-z}{w-\frac{t}{n}}} u^{\frac{1}{2}} (1-u)^{-\frac{1}{2}} \left\{ \left[\left(\left(w - \frac{t}{n} \right) (1-u) + \frac{t}{n} - z \right)^{\frac{1}{2}} \right. \right. \right. \\
&\quad \left. \left. - (w(1-u) - z)^{\frac{1}{2}} - \left(\left(w - \frac{t}{n} \right) (1-u) \right)^{\frac{1}{2}} + (w(1-u))^{\frac{1}{2}} \right] \frac{w - \frac{t}{n}}{z - \frac{t}{n}} \right. \right. \\
&\quad \left. \left. + \left[\left(\left(w - \frac{t}{n} \right) (1-u) + \frac{t}{n} - z \right)^{\frac{1}{2}} - \left(\left(w - \frac{t}{n} \right) (1-u) \right)^{\frac{1}{2}} \right] \frac{t(w-z)}{z(z - \frac{t}{n})} \right\} du \right| \\
&\leq C \left\{ \frac{1}{\sqrt{n}} \int_{-1+s}^{1-s} |\psi(t)| \left(\frac{nw-t}{nz-t} + \frac{w-z}{z\sqrt{nz-t}} \right) dt + \frac{1}{n\sqrt{z}} \int_{-1}^1 |\psi(t)| \frac{w-z}{w-\frac{t}{n}} dt \right\} \\
&\leq C \{ n^{-\frac{1}{2}} w z^{-1} + n^{-1} z^{-\frac{1}{2}} \} \leq C \{ n^{-\frac{1}{2}} (w-z) z^{-1} + n^{-\frac{1}{2}} \} \\
&\leq C \{ n^{\frac{1}{2}} (nz)^{-1} (w-z) + n^{-\frac{1}{2}} \},
\end{aligned}$$

also,

$$\begin{aligned}
|F_2^n(w, z)| &= \int_{-1}^1 |\psi(t)| dt \int_s^w (w-s)^{\frac{1}{2}} \left(s - \frac{t}{n} \right)^{-\frac{1}{2}} ds \int_0^1 \left(s - \frac{t}{n} + \theta \left(\frac{t}{n} - z \right) \right)^{-\frac{1}{2}} d\theta \\
&= (w-z)^{\frac{1}{2}} \int_{-1}^1 \frac{|\psi(t)|}{z - \frac{t}{n}} dt \int_s^w \left(\left(\frac{s - \frac{t}{n}}{s-z} \right)^{\frac{1}{2}} - 1 \right) ds \\
&= \frac{1}{2} (w-z)^{\frac{1}{2}} \int_{-1}^1 |\psi(t)| dt \int_0^{\frac{w-z}{z - \frac{t}{n}}} du \int_0^1 (1+\tau)^{-\frac{1}{2}} d\tau \\
&= \frac{1}{2} (w-z)^{\frac{1}{2}} \int_{-1}^1 |\psi(t)| \left[\int_{\frac{z - \frac{t}{n}}{w-z}}^{\infty} \frac{d\tau}{\tau(1+\tau)^{\frac{1}{2}}} + \int_0^{\frac{z - \frac{t}{n}}{w-z}} (1+\tau)^{-\frac{1}{2}} d\tau \int_0^{\frac{w-z}{z - \frac{t}{n}}} du \right] dt
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2}(w-z)^{\frac{1}{2}} \int_{-1}^1 |\psi(t)| \left[\int_{\frac{w-t}{w-z}}^{\frac{z-t}{w-z}} \frac{d\tau}{\tau(1+\tau)^{\frac{1}{2}}} + \frac{w-z}{z-\frac{t}{n}} \int_0^{\frac{z-t}{w-z}} (1+\tau)^{-\frac{1}{2}} d\tau \right] dt \\
&\leq \frac{1}{2}(w-z)^{\frac{1}{2}} \int_{-1}^1 |\psi(t)| \left[\left| \int_{\frac{w-t}{w-z}}^{\frac{z-t}{w-z}} \frac{1}{\tau} d\tau \right| + \int_1^{\infty} \frac{d\tau}{\tau(1+\tau)^{\frac{1}{2}}} + 1 \right] dt \\
&\leq O(w-z)^{\frac{1}{2}} \left[\int_{-1}^1 |\psi(t)| \left| \ln \frac{z-\frac{t}{n}}{w-z} \right| dt + 1 \right] \\
&\leq O(w-z)^{\frac{1}{2}} \left[\int_{-1}^1 |\psi(t)| \left| \ln \left(z - \frac{t}{n} \right) \right| dt + 1 \right] \leq O n^{\frac{1}{4}}.
\end{aligned}$$

So

$$|F_n(w, z)| \leq O \min(n^{\frac{1}{4}}, n^{\frac{1}{2}}(nz)^{-1}(w-z) + n^{-\frac{1}{2}}). \quad (2.39)$$

Furthermore,

$$\begin{aligned}
\left| \int_s^w F_1^n(w, y) dy \right| &\leq \frac{2}{3} \left\{ \left| \int_{-1}^1 \psi(t) \left[\int_0^{\frac{w-s}{w-\frac{t}{n}}} u^{\frac{1}{2}}(1-u)^{-\frac{1}{2}} \left(w-z - \left(w-\frac{t}{n} \right) u \right)^{\frac{3}{2}} du \right. \right. \right. \\
&\quad \left. \left. - \int_0^{\frac{w-s}{w}} u^{\frac{1}{2}}(1-u)^{-\frac{1}{2}} (w-z-wu)^{\frac{3}{2}} du \right] dt \right| + 3 \left| \int_{-1}^1 \psi(t) dt \right. \\
&\quad \times \int_s^w (w-s)^{-\frac{1}{2}}(s-z) ds \int_0^1 d\theta \int_0^1 \left[\left(s-\frac{t}{n} \right)^{-\frac{1}{2}} \left(\left(s-\frac{t}{n} \right) \theta + (s-z)(1-\theta)\tau \right)^{\frac{3}{2}} \right. \\
&\quad \left. \left. - s^{\frac{1}{2}}(s\theta + (s-z)(1-\theta)\tau)^{\frac{1}{2}} \right] d\tau \right| + 4(w-z)^{\frac{3}{2}} \left| \int_{-1}^1 \psi(t) dt \right| \Big\} \\
&\leq \frac{2}{3} \left\{ \frac{3}{2n} \int_{-1}^1 |t\psi(t)| dt \int_0^{\frac{w-s}{w-\frac{t}{n}}} u^{\frac{1}{2}}(1-u)^{-\frac{1}{2}} du \int_0^1 \left(w-z-wu + \theta u \frac{t}{n} \right)^{\frac{1}{2}} d\theta \right. \\
&\quad + \int_{-1}^1 |\psi(t)| dt \int_{\frac{w-s}{w}}^{\frac{w-s}{w-\frac{t}{n}}} u^{\frac{1}{2}}(1-u)^{-\frac{1}{2}} (w(1-u)-z)^{\frac{3}{2}} du \\
&\quad + 3 \left| \int_{-1}^1 \psi(t) dt \int_s^w (w-s)^{-\frac{1}{2}}(s-z) \left(s-\frac{t}{n} \right)^{-\frac{1}{2}} ds \int_0^1 d\theta \right. \\
&\quad \times \int_0^1 \left[\frac{\left(\left(s-\frac{t}{n} \right) \theta + (s-z)(1-\theta)\tau \right)^{\frac{3}{2}} - (s\theta + (s-z)(1-\theta)\tau)^{\frac{1}{2}}}{\left(\frac{|t|}{n} \theta \right)^{\frac{1}{2}}} \left(\frac{|t|}{n} \theta \right)^{\frac{1}{2}} \right. \\
&\quad \left. \left. - \frac{\left(s-\frac{t}{n} \right)^{\frac{1}{2}} - s^{\frac{1}{2}}}{\left(\frac{|t|}{n} \right)^{\frac{1}{2}}} \left(\frac{|t|}{n} \right)^{\frac{1}{2}} \left(\theta + \frac{s-z}{s}(1-\theta)\tau \right)^{\frac{1}{2}} \right] d\tau \right| \Big\} \\
&\leq O \left\{ \frac{w^{\frac{1}{2}}}{n} \int_{-1}^1 |t\psi(t)| dt \int_0^{\frac{w-s}{w-\frac{t}{n}}} u^{\frac{1}{2}} du + w^{\frac{1}{2}} \int_{-1}^1 \frac{w-z}{w-\frac{t}{n}} \frac{|t\psi(t)|}{n} dt \right\}
\end{aligned}$$

$$\begin{aligned}
& + \frac{w-z}{\sqrt{n}} \int_{\frac{t}{n}}^w (w-s)^{-\frac{1}{2}} \left(s - \frac{t}{n}\right)^{-\frac{1}{2}} ds \Big\} \\
& \leq C \left\{ \frac{1}{\sqrt{n}} \int_{-1+s}^{1-s} \frac{(w-z)}{\sqrt{nw-t} \sqrt{1-\frac{t}{nw}}} dt + \frac{w-z}{\sqrt{n(nw-t)}} \right. \\
& \quad \times \left. \int_{-1}^1 \frac{|\psi(t)|}{\left(1-\frac{t}{nw}\right)^{\frac{1}{2}}} dt + \frac{w-z}{\sqrt{n}} \right\} \leq C n^{-\frac{1}{2}} (w-z). \\
& \left| \int_s^w F_2^n(w, y) dy \right| = 2 \left| \int_{-1}^1 \psi(t) dt \int_s^w (w-s)^{\frac{1}{2}} \left(s - \frac{t}{n}\right)^{-\frac{1}{2}} ds \right. \\
& \quad \times \left. \int_0^1 \frac{\left(\left(s - \frac{t}{n}\right)(1-\theta)\right)^{\frac{1}{2}} - \left(s - \frac{t}{n} + \theta\left(\frac{t}{n} - z\right)\right)^{\frac{1}{2}}}{\theta} d\theta \right| \\
& = 2 \left| \int_{-1}^1 \psi(t) dt \int_s^w (w-s)^{\frac{1}{2}} ds \int_0^1 \frac{(1-\theta)^{\frac{1}{2}} - \left(1 + \theta \frac{\frac{t}{n} - z}{s - \frac{t}{n}}\right)^{\frac{1}{2}}}{\theta} d\theta \right| \\
& = 2 \left| \int_{-1}^1 \psi(t) dt \int_s^w (w-s)^{\frac{1}{2}} ds \int_0^1 \frac{\left(1 + \theta \frac{\frac{t}{n} - z}{s - \frac{t}{n}}\right)^{\frac{1}{2}} - \left(1 - \theta \frac{z}{s}\right)^{\frac{1}{2}}}{\theta} d\theta \right| \\
& \leq \frac{2}{\sqrt{n}} \int_{-1}^1 |\psi(t)| dt \int_s^w \frac{(w-s)^{\frac{1}{2}} (s-z)^{\frac{1}{2}}}{\left(s\left(s - \frac{t}{n}\right)\right)^{\frac{1}{2}}} ds \\
& \quad \times \int_0^1 \left| \frac{\left(1 + \theta \frac{\frac{t}{n} - z}{s - \frac{t}{n}}\right)^{\frac{1}{2}} - \left(1 - \theta \frac{z}{s}\right)^{\frac{1}{2}}}{\left(\frac{\frac{t}{n}(s-z)}{s\left(s - \frac{t}{n}\right)}\right)^{\frac{1}{2}}} \right| \frac{d\theta}{\theta^{\frac{1}{2}}} \\
& \leq \frac{C}{\sqrt{n}} \int_{-1}^1 |\psi(t)| dt \int_s^w (w-s)^{\frac{1}{2}} (s-z)^{-\frac{1}{2}} ds \leq \frac{C(w-z)}{\sqrt{n}}.
\end{aligned}$$

Hence

$$\left| \int_s^w F_n(w, y) dy \right| \leq C \frac{w-z}{\sqrt{n}}. \quad (2.40)$$

Observe that

$$\begin{aligned}
\left| (w-z)^{\frac{3}{2}} \int_{nw}^{\infty} \tilde{\psi}(y) dy \right| &= (w-z)^{\frac{3}{2}} \left| \int_{-1+s}^{1-s} t\psi(t) dt \int_{nw}^{\infty} \frac{dy}{y(y-t)} \right| \\
&\leq (w-z)^{\frac{3}{2}} \int_{-1+s}^{1-s} |t\psi(t)| dt \int_{nw}^{\infty} \frac{dy}{(y-1+s)^2} \\
&\leq O \frac{(w-z)^{\frac{3}{2}}}{nw-1+s} \leq O \frac{w-z}{\sqrt{n}} \sqrt{\frac{1-\frac{z}{w}}{\left(1-\frac{1-s}{nw}\right)(nw-1+s)}} \\
&\leq O \frac{w-z}{\sqrt{n}}, \\
\left| \int_{ns}^{nw} \left[(w-z)^{\frac{3}{2}} - \left(w-\frac{y}{n}\right)^{\frac{3}{2}} \right] \tilde{\psi}(y) dy \right| \\
&= \left| \int_{-1}^1 t\psi(t) dt \int_{ns}^{nw} \frac{(w-z)^{\frac{3}{2}} - \left(w-\frac{y}{n}\right)^{\frac{3}{2}}}{y(y-t)} dy \right| \\
&\leq (w-z)^{\frac{1}{2}} \left| \int_{-1}^1 t\psi(t) dt \int_{ns}^{nw} \frac{\frac{y}{n} - z}{y(y-t)} dt \right| \\
&\leq (w-z)^{\frac{1}{2}} \int_{-1}^1 |t\psi(t)| dt \int_{ns}^{nw} \frac{\left(\frac{1}{n} - \frac{z}{y}\right)^{\frac{1}{2}}}{(y-1+s)^{\frac{3}{2}}} dy \leq O n^{-\frac{1}{2}} (w-z) \int_{ns}^{nw} \frac{dy}{(y-1+s)^{\frac{3}{2}}} \\
&\leq O \frac{w-z}{\sqrt{n}}.
\end{aligned}$$

This shows that (2.38) holds.

§ 3. Convergence Theorem

Before a detailed discussion, we first introduce an important lemma.

Lemma 2. Suppose a sequence of approximate solutions $v^l(x, t) = (\rho^l(x, t), m^l(x, t)) = (\rho^l(x, t), \rho^l(x, t)u^l(x, t))$ of (1.1) ($1 < \gamma \leq 2$) satisfies the compactness frame (see [1]), that is,

(i) There exists a constant $O > 0$ such that

$$0 \leq \rho^l(x, t) \leq O, \quad |u^l(x, t)| \leq O.$$

(ii) The measure

$$\eta(\rho^l, m^l)_t + q(\rho^l, m^l)_s$$

lies in a compact subset of H_{loc}^{-1} for all weak entropy pair (η, q) . Then, on any bounded domain Ω , there is a family of probability measures $\{\nu_{s,t}\}_{(s,t) \in \Omega}$, $\text{supp } \nu_{s,t} \subset K$ (K is a bounded set containing v^l), which is uniquely determined by $v^l(x, t)$, such that, for any continuous or bounded

measurable weak-entropy flux pairs (η_1, q_1) and (η_2, q_2) , the commutativity relation (for simplicity, drop the indices x, t in $\nu_{x,t}$)

$$\langle \nu, \eta_1 \rangle \langle \nu, q_2 \rangle - \langle \nu, \eta_2 \rangle \langle \nu, q_1 \rangle = \langle \nu, \eta_1 q_2 - \eta_2 q_1 \rangle \quad (3.1)$$

holds.

The proof of the lemma may be found in the paper [4] or [5]. Details are left to readers.

Now we start from the commutativity relation (3.1) and prove the convergence theorem by using the results of § 2.

Convergence theorem. Suppose a sequence of approximate solutions $v^l(x, t) = (\rho^l(x, t), m^l(x, t))$ satisfies the conditions of Lemma 2, then there is a convergent subsequence such that

$$(\rho^{l_k}(x, t), m^{l_k}(x, t)) \rightarrow (\rho(x, t), m(x, t)). \quad \text{a.e.} \quad (3.2)$$

Set $V = \{(\rho, v): \rho = 0\} = \{(w, z): w = z\}$, K_0 is a minimal triangle which contains $\text{supp } \nu \setminus V$ and P is the vertex of the triangle $K_0 \setminus V$, then from the discussion in the paper [4], it suffices to prove that

$$\lim_{n \rightarrow \infty} \{\langle \nu|_{S_w^e}, n(w-z)^3 \rangle + \langle \nu|_{S_z^e}, n(w-z)^3 \rangle\} = 0, \quad (3.3)$$

where

$$S_w^e = \{(w, z): |w| < \frac{1-3\varepsilon}{n}\}, \quad S_z^e = \{(w, z): |z| < \frac{1-3\varepsilon}{n}\},$$

ε small constant.

To do this end, we first prove the following several lemmata.

Lemma 3. If $\text{supp } \nu \setminus V \neq \emptyset$, then $P \in \text{supp } \nu$.

This lemma is a result of the paper [4].

Lemma 4. For the weak-entropy pairs (η_n, q_n) gained in (B) of § 2, the expected values $\langle \nu, \eta_n \rangle$ and $\langle \nu, q_n \rangle$ are bounded in n . It immediately follows that $\langle \nu, \eta_n \hat{q}_n - \hat{\eta}_n q_n \rangle$ is bounded in n .

Similar to the paper [4], it is easy to prove the result by using A-3 of § 2, Lemma 1 and Lemma 3.

Lemma 5.

$$\lim_{n \rightarrow \infty} \langle \nu, \eta_n \hat{q}_n - \hat{\eta}_n q_n \rangle = 0. \quad (3.4)$$

Proof. Suppose, on the contrary that there is a subsequence such that

$$\lim_{k \rightarrow \infty} \langle \nu, \eta_{n_k} \hat{q}_{n_k} - \hat{\eta}_{n_k} q_{n_k} \rangle \neq 0, \quad (3.5)$$

we may extract a subsequence (we suppose $\{n_k\}$ itself) such that all limits of the sequences $\{\langle \nu, \eta_{n_k} \rangle\}$, $\{\langle \nu, q_{n_k} \rangle\}$, $\{\langle \nu, \hat{\eta}_{n_k} \rangle\}$ and $\{\langle \nu, \hat{q}_{n_k} \rangle\}$ exist.

If $\lim_{k \rightarrow \infty} \langle \nu, \eta_{n_k} \rangle = 0$ and/or $\lim_{k \rightarrow \infty} \langle \nu, \hat{\eta}_{n_k} \rangle = 0$, without loss of generality we suppose that $\lim_{k \rightarrow \infty} \langle \nu, \eta_{n_k} \rangle = 0$, then, using the equality

$$\langle \nu, \eta_0 q_{n_k} - \eta_{n_k} q_0 \rangle = \langle \nu, \eta_0 \rangle \langle \nu, q_{n_k} \rangle - \langle \nu, \eta_{n_k} \rangle \langle \nu, q_0 \rangle,$$

we have

$$\langle \nu, \eta_0 \rangle \lim_{k \rightarrow \infty} \langle \nu, q_{n_k} \rangle = 0.$$

We remark from the expression of η_0 and Lemma 3 that $\langle \nu, \eta_0 \rangle > 0$ and hence

$$\lim_{k \rightarrow \infty} \langle \nu, q_{n_k} \rangle = 0.$$

Therefore,

$$\begin{aligned} |\langle \nu, \eta_{n_k} \hat{q}_{n_k} - \hat{\eta}_{n_k} q_{n_k} \rangle| &= |\langle \nu, \eta_{n_k} \rangle \langle \nu, \hat{q}_{n_k} \rangle - \langle \nu, \hat{\eta}_{n_k} \rangle \langle \nu, q_{n_k} \rangle| \\ &\leq O(|\langle \nu, \eta_{n_k} \rangle| + |\langle \nu, q_{n_k} \rangle|) \rightarrow 0, \quad (k \rightarrow \infty). \end{aligned}$$

This violates (3.5).

If neither $\lim_{k \rightarrow \infty} \langle \nu, \eta_{n_k} \rangle$ nor $\lim_{k \rightarrow \infty} \langle \nu, \hat{\eta}_{n_k} \rangle$ is zero, then, from the commutativity relations

$$\begin{cases} \langle \nu, \eta_0 \rangle \frac{\langle \nu, q_{n_k} \rangle}{\langle \nu, \eta_{n_k} \rangle} - \langle \nu, q_0 \rangle = \frac{\langle \nu, \eta_0 q_{n_k} - \eta_{n_k} q_0 \rangle}{\langle \nu, \eta_{n_k} \rangle}, \\ \langle \nu, \eta_0 \rangle \frac{\langle \nu, \hat{q}_{n_k} \rangle}{\langle \nu, \hat{\eta}_{n_k} \rangle} - \langle \nu, q_0 \rangle = \frac{\langle \nu, \eta_0 \hat{q}_{n_k} - \hat{\eta}_{n_k} q_0 \rangle}{\langle \nu, \hat{\eta}_{n_k} \rangle}. \end{cases}$$

We have

$$\begin{cases} \langle \nu, \eta_0 \rangle \lim_{k \rightarrow \infty} \frac{\langle \nu, q_{n_k} \rangle}{\langle \nu, \eta_{n_k} \rangle} - \langle \nu, q_0 \rangle = 0, \\ \langle \nu, \eta_0 \rangle \lim_{k \rightarrow \infty} \frac{\langle \nu, \hat{q}_{n_k} \rangle}{\langle \nu, \hat{\eta}_{n_k} \rangle} - \langle \nu, q_0 \rangle = 0. \end{cases}$$

Since $\langle \nu, \eta_0 \rangle > 0$, we obtain

$$\lim_{k \rightarrow \infty} \left(\frac{\langle \nu, q_{n_k} \rangle}{\langle \nu, \eta_{n_k} \rangle} - \frac{\langle \nu, \hat{q}_{n_k} \rangle}{\langle \nu, \hat{\eta}_{n_k} \rangle} \right) = 0.$$

Thus

$$\begin{aligned} |\langle \nu, \eta_{n_k} \hat{q}_{n_k} - \hat{\eta}_{n_k} q_{n_k} \rangle| &= \left| \langle \nu, \eta_{n_k} \rangle \langle \nu, \hat{\eta}_{n_k} \rangle \left(\frac{\langle \nu, \hat{q}_{n_k} \rangle}{\langle \nu, \hat{\eta}_{n_k} \rangle} - \frac{\langle \nu, q_{n_k} \rangle}{\langle \nu, \eta_{n_k} \rangle} \right) \right| \\ &\leq O \left| \frac{\langle \nu, \hat{q}_{n_k} \rangle}{\langle \nu, \hat{\eta}_{n_k} \rangle} - \frac{\langle \nu, q_{n_k} \rangle}{\langle \nu, \eta_{n_k} \rangle} \right| \rightarrow 0, \quad (k \rightarrow \infty). \end{aligned}$$

This arrives at a contradiction which is similar to the above one. The result follows.

Lemma 6. The probability measures $\nu_{a,i}$ satisfy

$$\begin{aligned} &\langle \nu|_{s_w}, n(w-z)^3 A_1(nw) + (w-z)^{\frac{3}{2}} A_2(w, z) \rangle + \langle \nu|_{s_z}, n(w-z)^3 B_1(nz) \\ &\quad + (w-z)^{\frac{3}{2}} B_2(w, z) \rangle + \langle \nu|_{q_1}, n(w-z)^{\frac{3}{2}} [C_1(nz)(w-z)^{\frac{3}{2}} + C_2(w, z)] \\ &\quad + (w-z)^{\frac{3}{2}} C_3(w, z) \rangle + \langle \nu|_{u_1}, n(w-z)^3 D_1(nz) \\ &\quad + (w-z)^{\frac{3}{2}} D_2(w, z) \rangle \rightarrow 0, \quad (n \rightarrow \infty), \end{aligned} \quad (3.6)$$

where

$$S_w = \left\{ (w, z) : |w| \leq \frac{1}{n} \right\}, \quad S_z = \left\{ (w, z) : |z| \leq \frac{1}{n} \right\},$$

$$Q_1 = \left\{ (w, z) : w > \frac{1}{n}, z < -\frac{1}{n}, w > -Lz \right\}, \quad U_2 = \left\{ (w, z) : w > \frac{1}{n}, z > \frac{1}{n} \right\}.$$

Proof. We notice that

$$\langle \nu, B_n \rangle = \langle \nu |_{S_w \cup S_z \cup Q_1 \cup U_2}, B_n \rangle + \langle \nu |_{Q_2 \cup U_1}, B_n \rangle$$

and recall that

$$|\langle \nu |_I, B_n \rangle| + |\langle \nu |_{Q_2 \cup U_1}, B_n \rangle| \rightarrow 0, \quad (n \rightarrow \infty).$$

Hence we obtain from Lemma 5 that

$$\lim_{n \rightarrow \infty} \langle \nu |_{S_w \cup S_z \cup Q_1 \cup U_2}, B_n \rangle = \lim_{n \rightarrow \infty} (\langle \nu, B_n \rangle - \langle \nu |_{Q_2 \cup U_1}, B_n \rangle) = 0.$$

Using the expressions and estimates of B_n on the various domains in (D) of § 2, we may immediately obtain the result.

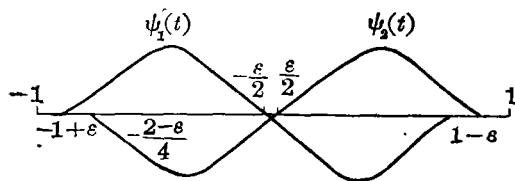


Fig. 2

Up to now, we can prove the theorem by using the above lemmata.

Proof of the theorem.

It suffices to prove (3.3) by virtue of the above discussion. Define

$$\begin{aligned} \psi_1(t) &= a_{2-3\epsilon} \left(t + \frac{2-\epsilon}{4} \right) - a_{2-3\epsilon} \left(t - \frac{2-5\epsilon}{4} \right), \\ \psi_2(t) &= -a_{2-3\epsilon} \left(t + \frac{2-5\epsilon}{4} \right) + a_{2-3\epsilon} \left(t - \frac{2-\epsilon}{4} \right), \end{aligned}$$

where $a_r(t-t_0) = ra\left(\frac{t-t_0}{r}\right)$ and

$$a(t) = \begin{cases} \frac{1}{e^{|t|}-1}, & |t| < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$\left. \begin{aligned} \psi_i(t) &\in C_n^\infty(-1, 1), \text{ supp } \psi_i(t) \subset (-1+\epsilon, 1-\epsilon), \quad \int_{-1}^1 \psi_i(s) ds = 0, \\ \left(\int_{-1}^t \psi_i(s) ds \right)^2 &\geq C_\epsilon^2 > 0, \text{ on } [-1+3\epsilon, 1-3\epsilon], \quad (i=1, 2), \end{aligned} \right\} \quad (3.7)$$

and

$$\psi_1\left(t - \frac{\epsilon}{2}\right) = -\psi_2\left(t + \frac{\epsilon}{2}\right).$$

Also,

$$\int_{-\infty}^x \tilde{\psi}_1(y) dy = \int_{-1}^1 \psi_1(t) \ln |x-t| dt = \int_0^1 \psi_1\left(t - \frac{s}{2}\right) \ln \left| \frac{x - \left(t - \frac{s}{2}\right)}{x + \left(t + \frac{s}{2}\right)} \right| dt.$$

$$\begin{aligned} \int_{-\infty}^x \tilde{\psi}_2(y) dy &= \int_0^1 \psi_2\left(t + \frac{s}{2}\right) \ln \left| \frac{x - \left(t + \frac{s}{2}\right)}{x + t - \frac{s}{2}} \right| dt \\ &\quad - \int_0^1 \psi_1\left(t - \frac{s}{2}\right) \ln \left| \frac{x - \left(t + \frac{s}{2}\right)}{x + t - \frac{s}{2}} \right| dt. \end{aligned}$$

It is obvious that the functions $\int_{-\infty}^x \tilde{\psi}_1(y) dy$ and $\int_{-\infty}^x \tilde{\psi}_2(y) dy$ have not a common zero point on any bounded interval, that is,

$$\left(\int_{-\infty}^x \tilde{\psi}_1(y) dy\right)^2 + \left(\int_{-\infty}^x \tilde{\psi}_2(y) dy\right)^2 \geq C_N > 0, \quad x \in [-N, N]. \quad (3.8)$$

Define $\hat{\psi}_i(t) = \int_{-1}^t \psi_i(s) ds + t\psi_i(t)$ ($i=1, 2$). Then $\hat{\psi}_i(t)$ ($i=1, 2$) have the same properties as $\psi_i(t)$ in (3.7) and

$$\begin{aligned} A_1^4(nw) &= \pi^2 \left(\int_{-1}^{nw} \psi_i(t) dt \right)^2 \geq \pi^2 C_0^2 > 0, \quad nw \in [-1+3s, 1-3s], \\ B_1^1(nz) + B_1^2(nz) &= \left(\int_{-\infty}^{nz} \tilde{\psi}_1(y) dy \right)^2 + \left(\int_{-\infty}^{nz} \tilde{\psi}_2(y) dy \right)^2 \geq C_0 > 0, \quad nz \in [-1, 1], \end{aligned} \quad (3.9)$$

($i=1, 2$) correspond to $\psi(t) = \psi_i(t)$.

Now we consider

$$\langle \nu|_{S_w}, n(w-z)^{\frac{3}{2}} A_1^4(nw) + (w-z)^{\frac{3}{2}} A_2^4(w, z) \rangle, \quad (i=1, 2),$$

where

$$A_1^4(nw) = \pi^2 \left(\int_{-1}^{nw} \psi_i(t) dt \right)^2 \text{ and } |A_2^4(w, z)| \leq O\left(|\psi_i(nw)| + \left| \int_{-1}^{nw} \psi_i(t) dt \right| \right),$$

$$(i=1, 2).$$

We obtain from the character of the function ψ_1 that there is a constant sequence $\{\delta_n\}$ ($\delta_n > s$, $\delta_n - s \rightarrow 0$ ($n \rightarrow \infty$)) such that

$$\text{If } w \in \left[\frac{-1+\delta_n}{n}, \frac{1-\delta_n-s}{n} \right] \equiv S_{w^n}^s, \text{ then } \left(\int_{-1}^{nw} \psi_1(t) dt \right)^2 \geq \frac{1}{\sqrt{n}}.$$

$$\text{If } w \in \left[-\frac{1}{n}, \frac{-1+\delta_n}{n} \right] \cup \left[\frac{1-s-\delta_n}{n}, \frac{1}{n} \right] \equiv S_w - S_{w^n}^s, \text{ then } |\psi_1(t)| \leq o(1).$$

Set

$$A_n = \{(w, z): w - z \leq n^{-\frac{1}{4}}\}.$$

when $w \in S_w - S_w^{\delta_n}$, $|(w-z)^3 A_2^1(w, z)| \leq o(1)$,

when $w \in S_w^{\delta_n}$, $\left(\int_{-1}^{nw} \psi_1(t) dt\right)^2 \geq n^{-\frac{1}{2}}$.

If $(w, z) \in A_n$, then

$$|(w-z)^{\frac{3}{2}} A_2^1(w, z)| \leq C n^{-\frac{3}{8}}.$$

If $(w, z) \in A_n^c$, then for large n ,

$$\begin{aligned} |(w-z)^{\frac{3}{2}} A_2^1(w, z)| &\leq (w-z)^3 \frac{C}{(w-z)^{\frac{3}{2}}} \leq C n^{\frac{3}{8}} (w-z)^3 \leq \frac{\sqrt{n}}{\pi^2} (w-z)^3 \\ &\leq n(w-z)^3 A_1^1(nw). \end{aligned}$$

Hence

$$\begin{aligned} &\langle \nu|_{S_w}, n(w-z)^3 A_1^1(nw) + (w-z)^{\frac{3}{2}} A_2^1(w, z) \rangle \\ &= \langle \nu|_{S_w \cap A_n}, n(w-z)^3 A_1^1(nw) \rangle + \langle \nu|_{S_w \cap A_n^c}, n(w-z)^3 A_1^1(nw) \\ &\quad + (w-z)^{\frac{3}{2}} A_2^1(w, z) \rangle + \langle \nu|_{S_w - S_w^{\delta_n}}, n(w-z)^3 A_1^1(nw) \rangle + o(1) \\ &\geq \langle \nu|_{S_w \cap A_n}, n(w-z)^3 A_1^1(nw) \rangle + \langle \nu|_{S_w \cap A_n^c}, n(w-z)^3 A_1^1(nw) \\ &\quad + (w-z)^{\frac{3}{2}} A_2^1(w, z) \rangle + o(1) \\ &\geq O_s(\langle \nu|_{S_w \cap A_n}, n(w-z)^3 \rangle + \langle \nu|_{S_w \cap A_n^c}, n(w-z)^3 (1 - C n^{-\frac{5}{8}}) \rangle) + o(1) \\ &\geq \alpha_0 \langle \nu|_{S_w}, n(w-z)^3 \rangle + o(1), \quad (n \text{ sufficiently large}). \end{aligned} \quad (3.10)$$

For $\psi_2(x)$, we can similarly obtain the same result.

Next, we consider

$$\langle \nu|_{S_z}, n(w-z)^3 \sum_{i=1}^2 B_i^1(nz) + (w-z)^{\frac{3}{2}} \sum_{i=1}^2 B_i^1(w, z) \rangle,$$

where $B_1^1(nz) = \left(\int_{-1}^{nz} \tilde{\psi}_1(y) dy\right)^2$ and $|B_2^1(w, z)| \leq C[n^{\frac{1}{2}}(w-z) + n^{-\frac{1}{2}}]$.

We know from (3.9) that

$$\sum_{i=1}^2 B_i^1(nz) \geq C_0 > 0, \quad |nz| \leq 1.$$

Thus

$$\begin{aligned} &\langle \nu|_{S_z}, n(w-z)^3 \sum_{i=1}^2 B_i^1(nz) + (w-z)^{\frac{3}{2}} \sum_{i=1}^2 B_i^1(w, z) \rangle \\ &\geq C_0 \langle \nu|_{S_z}, n(w-z)^3 \rangle - C n^{\frac{1}{2}} (w-z)^{\frac{5}{2}} + o(1). \end{aligned}$$

If $(w, z) \in A_n$, then

$$n^{\frac{1}{2}} (w-z)^{\frac{5}{2}} \leq n^{-\frac{1}{8}}.$$

If $(w, z) \in A_n^c$, then for large n ,

$$C n^{\frac{1}{2}} (w-z)^{\frac{5}{2}} = C n^{\frac{1}{2}} (w-z)^{-\frac{1}{2}} (w-z)^3 \leq C n^{\frac{5}{8}} (w-z)^3 < n(w-z)^3.$$

So

$$\begin{aligned}
 & \langle \nu|_{S_2}, n(w-z)^3 \sum_{i=1}^2 B_1^i(nz) + (w-z)^{\frac{3}{2}} \sum_{i=1}^2 B_2^i(w, z) \rangle \\
 & \geq C(\langle \nu|_{S_2 \cap A_n}, n(w-z)^3 \rangle + \langle \nu|_{S_2 \cap A_n^c}, n(w-z)^3(1 - Cn^{-\frac{5}{8}}) \rangle) + o(1) \\
 & \geq \alpha_0(\langle \nu|_{S_2 \cap A_n}, n(w-z)^3 \rangle + \langle \nu|_{S_2 \cap A_n^c}, n(w-z)^3 \rangle) + o(1) \\
 & \geq \alpha_0 \langle \nu|_{S_2}, n(w-z)^3 \rangle + o(1). \quad (3.11)
 \end{aligned}$$

Furthermore, we have

$$\begin{aligned}
 & \langle \nu|_{Q_1}, n(w-z)^{\frac{3}{2}} [C_1^i(nz)(w-z)^{\frac{3}{2}} + C_2^i(w, z)] + (w-z)^{\frac{3}{2}} C_3^i(w, z) \rangle \\
 & = \langle \nu|_{Q_1}, nC_1^i(nz)(w-z)^{\frac{3}{2}} \left[(w-z)^{\frac{3}{2}} + \frac{C_2^i(w, z)}{C_1^i(nz)} \right] + (w-z)^{\frac{3}{2}} C_3^i(w, z) \rangle \\
 & \geq \langle \nu|_{Q_1}, C_1^i(nz) [nC_0(w-z)^3 - Cn^{\frac{1}{2}}(w-z)^{\frac{5}{2}}] \rangle.
 \end{aligned}$$

If $(w, z) \in A_n$, then

$$n^{\frac{1}{2}}(w-z)^{\frac{5}{2}} \leq n^{-\frac{1}{8}}.$$

If $(w, z) \in A_n^c$, then

$$Cn^{\frac{1}{2}}(w-z)^{\frac{5}{2}} \leq Cn^{\frac{5}{8}}(w-z)^3 < C_0 n(w-z)^3, \quad (n \text{ sufficiently large}).$$

Thus

$$\begin{aligned}
 & \langle \nu|_{Q_1}, n(w-z)^{\frac{3}{2}} [C_1^i(nz)(w-z)^{\frac{3}{2}} + C_2^i(w, z)] + (w-z)^{\frac{3}{2}} C_3^i(w, z) \rangle \\
 & \geq \langle \nu|_{Q_1 \cap A_n}, nC_0 C_1^i(nz)(w-z)^3 \rangle + o(1) \geq o(1), \quad (i=1, 2). \quad (3.12)
 \end{aligned}$$

Finally, we similarly have

$$\begin{aligned}
 & \langle \nu|_{U_1}, n(w-z)^3 D_1^i(nz) + (w-z)^{\frac{3}{2}} D_2^i(w, z) \rangle \\
 & = \langle \nu|_{U_1}, nD_1^i(nz)(w-z)^3 - Cn^{\frac{1}{2}}(w-z)^{\frac{5}{2}} D_1^i(nz) \rangle + o(1) \\
 & \geq \langle \nu|_{U_1}, D_1^i(nz) [n(w-z)^3 - Cn^{\frac{1}{2}}(w-z)^{\frac{5}{2}}] \rangle + o(1) \\
 & = \langle \nu|_{U_1 \cap A_n}, nD_1^i(nz)(w-z)^3 \rangle + \langle \nu|_{U_1 \cap A_n^c}, n(1 - n^{-\frac{3}{8}}) D_1^i(nz)(w-z)^3 \rangle + o(1) \\
 & \geq o(1), \quad (n \text{ sufficiently large}).
 \end{aligned}$$

From (3.6) and (3.10)–(3.13), we obtain

$$\lim_{n \rightarrow \infty} (\langle \nu|_{S_2}, n(w-z)^3 \rangle + \langle \nu|_{S_2^c}, n(w-z)^3 \rangle) = 0.$$

This completes the proof of the theorem. ■

Corollary. Assume that $\gamma = \frac{3}{2}$, then there exist a global generalized solution $(\rho(x, t), u(x, t))$ of Cauchy problem (1.1)–(1.2).

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